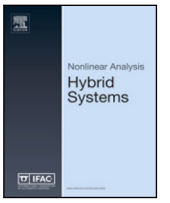
Nonlinear Analysis: Hybrid Systems 52 (2024) 101467



Contents lists available at ScienceDirect

Nonlinear Analysis: Hybrid Systems

journal homepage: www.elsevier.com/locate/nahs





Data-driven abstraction-based control synthesis ☆

Milad Kazemi [a], Rupak Majumdar [b], Mahmoud Salamati [b], Sadegh Soudjani [b,*], Ben Wooding [c]

[a] Department of Informatics, King's College London, United Kingdom
[b] Max Planck Institute for Software Systems, Kaiserslautern, Germany
[c] School of Computing, Newcastle University, United Kingdom

ARTICLE INFO

Keywords:
Abstraction-based methods
Data-driven synthesis
Formal controller synthesis
Sample complexity

ABSTRACT

This paper studies formal synthesis of controllers for continuous-space systems with unknown dynamics to satisfy requirements expressed as linear temporal logic formulas. Formal abstraction-based synthesis schemes rely on a precise mathematical model of the system to build a finite abstract model, which is then used to design a controller. The abstraction-based schemes are not applicable when the dynamics of the system are unknown. We propose a data-driven approach that computes a growth bound of the system using a finite number of trajectories. The computed growth bound together with the sampled trajectories are then used to construct the abstraction and synthesise a controller.

Our approach casts the computation of a growth bound as a robust convex optimisation program (RCP). Since the unknown dynamics appear in the optimisation, we formulate a scenario convex program (SCP) corresponding to the RCP using a finite number of sampled trajectories. We establish a sample complexity result that gives a lower bound for the number of sampled trajectories to guarantee the correctness of the growth bound computed from the SCP with a given confidence. Our sample complexity result requires knowing a possibly conservative bound on the Lipschitz constant of the system. We also provide a sample complexity result for the satisfaction of the specification on the system in closed loop with the designed controller for a given confidence. Our data-driven synthesised controller can provide guarantees on satisfaction of both finite and infinite-horizon specifications. We show that our data-driven approach can be readily used as a model-free abstraction refinement scheme by modifying the formulation of the system's growth bounds and providing similar sample complexity results. The performance of our approach is shown on three case studies.

1. Introduction

One of the major objectives in the design of safety-critical systems is to ensure their safe operation while satisfying high-level requirements. Examples of safety-critical systems include power grids, autonomous vehicles, traffic control, and battery-powered medical devices. Automatic design of controllers for such systems that can fulfil the given requirements have received significant attention recently. These systems can be represented as control systems with continuous state spaces. Within these continuous spaces,

☆ The authors are ordered alphabetically. The corresponding author was with the School of Computing, Newcastle University, United Kingdom, during the research of this paper. This work is supported by the EPSRC New Investigator Award CodeCPS (EP/V043676/1), the European Innovation Council project SymAware (101070802), the European Research Council (101089047), and the DFG, Germany project 389792660 TRR 248–CPEC.

* Corresponding author at: Max Planck Institute for Software Systems, Kaiserslautern, Germany.
E-mail address: sadegh@mpi-sws.org (S. Soudjani).

https://doi.org/10.1016/j.nahs.2024.101467
Received 19 June 2022; Received in revised form 1 November 2023; Accepted 20 January 2024
Available online 29 January 2024
1751-570X/© 2024 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

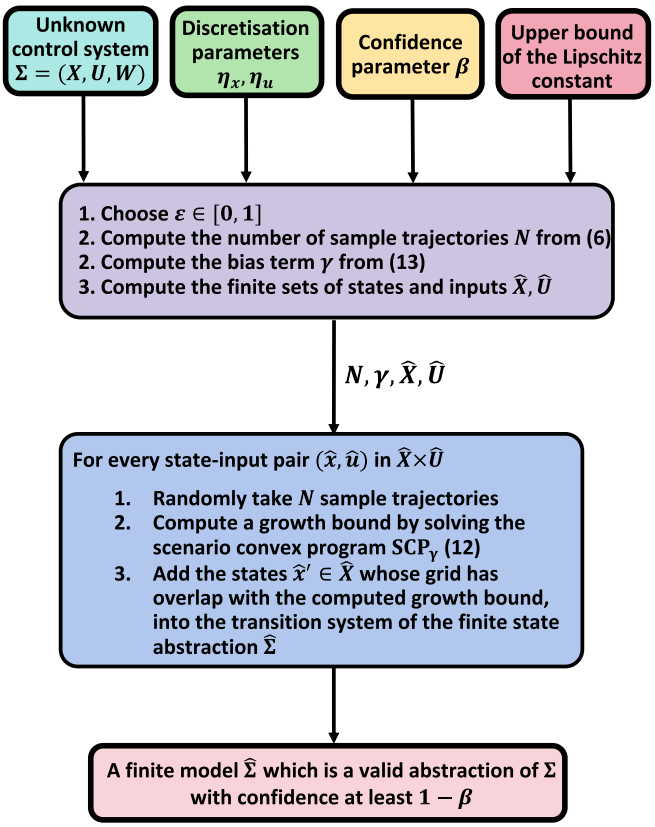


Fig. 1. A diagram illustrating the steps of the proposed data-driven method for constructing finite abstractions.

it is challenging to leverage automated control synthesis methods that provide satisfaction guarantees for high-level specifications, such as those expressed in linear temporal logic [1–4].

A common approach to tackle the continuous nature of the state space is to use abstraction-based controller design (ABCD) schemes [2,3,5,6]. The first step in the ABCD scheme is to compute a finite abstraction by discretising the state and action spaces. Finite abstractions are connected to the original system via an appropriate behavioural relation such as feedback refinement relations or alternating bisimulation relations [3,7]. Under such behavioural relations, trajectories of the abstraction are related to the ones of the original system. Therefore, a controller designed for the simpler finite abstract system can be refined to a controller for the original system. The controller designed by the ABCD scheme is described as being formal due to the guarantees on satisfaction of the specification by the original system in closed loop with the designed controller.

ABCD schemes generally rely on a precise mathematical model of the system. This stems from the fact that establishing a behavioural relation between the original system and its finite abstraction uses reachability analysis over the dynamics of the original system that require knowledge of the dynamical equations. Although such equations can in principle be derived for instance by using physics laws, the real-world control systems are a mixture of differential equations, block diagrams, and lookup tables. Therefore, extracting a clean analytical model for systems of practical interest could be infeasible. A promising approach to tackle this issue is to develop data-driven control synthesis schemes with appropriate formal (probabilistic) guarantees.

The main contribution of this paper is to provide a data-driven approach for formal synthesis of controllers to satisfy temporal specifications. A brief overview of the approach can be seen in Fig. 1. We focus on continuous-time nonlinear dynamical systems whose dynamics are unknown but sampled trajectories are available. Our approach constructs a *finite* abstract model of the system using only a finite number of sampled trajectories and a growth bound of the system. We formulate the computation of a growth bound as a robust convex program (RCP) that has infinite uncountable number of constraints. We then approximate the solution of the RCP with a scenario convex program (SCP) that has a finite number of constraints and can be solved using only a finite set of sampled trajectories. We establish a sample complexity result that gives a lower bound for the required number of trajectories to guarantee the correctness of the computed growth bound *over the whole state space* with a given confidence. We also provide a sample complexity result for the satisfaction of the specification on the system in a closed loop with the designed controller for a given confidence. As our last contribution, we show that our approach can be extended to a model-free abstraction refinement scheme by modifying the formulation of the system's growth bounds and providing similar sample complexity results. We demonstrate the performance of our approach on three case studies.

Our sample complexity result requires knowing a (possibly conservative) bound on the Lipschitz constant of the system. Algorithms founded on the extreme value theory can be utilised to *estimate* the true Lipschitz constant [8,9]. The estimated values will converge to a true Lipschitz constant when the number of samples used for this estimation goes to infinity. Our sample complexity result for the abstraction and synthesis is valid under the assumption that these estimations give a correct upper bound on the Lipschitz constant.

The remainder of this paper is organised as follows. After discussing the related work, Section 2 covers preliminaries on dynamical systems and finite abstractions and provides the problem statement. In Section 3, we present the assumptions and theoretical results needed for connecting RCPs and their corresponding SCPs. In Section 4, we present our approach on data-driven computation of a growth bound and the abstraction and prove our sample complexity result under the assumption of knowing a bound on the Lipschitz constant of the system. Estimation of a bound on the Lipschitz constant of the system for computing the number of samples is also discussed in this section. Section 5 discusses the extension of our approach to a data-driven abstraction refinement scheme. Several numerical examples are provided in Section 6 that support the theoretical findings of our paper. Finally, Section 7 contains concluding remarks and future research directions.

Related Work. There is an extensive body of literature on *model-based* formal synthesis for both deterministic and probabilistic systems. We refer the reader to the books [1–3], seminal papers [4,10], and the survey paper [11]. *Data-driven* approaches for analysis, verification, and synthesis of systems have received significant attention recently to improve efficiency and scalability of model-based approaches, and to study problems in which a model of the system is either not available or costly and time-consuming to construct. Given a prior inaccurate knowledge about the model of the system, a research line is to use data for refining the model and then synthesise a controller. Such approaches assume a class of models and improve the estimation of the uncertainty within the model class. These approaches range from using Gaussian processes [12,13], differential inclusions [14], rapidly-exploring random graphs [15], piecewise affine models [16], and model-based reinforcement learning algorithms [17]. A data-driven framework is proposed by Fan et al. [18] for verifying properties of hybrid systems when the continuous dynamics are unknown but the discrete transitions are known.

Data-driven approaches for solutions of scenario convex programs are developed for switching systems by Wang and Jungers to establish stability [19] and by Berger et al. for invariant subspace identification [20]. Ahmad et al. [21] have developed an adaptive sampling-based approach for motion planning using deterministic nonlinear control systems and robust control barrier functions. Zhong et al. [22] have studied linear dynamical systems with bounded disturbances by proposing a data-driven method to compute state feedback controllers that enforce staying in safety invariant sets by using finite number of state-input data points. Cohen et al. [23] have developed a model-based reinforcement learning approach to satisfy linear temporal logic specifications on continuous-time nonlinear systems.

Data-driven model-free approaches compute the solution of the synthesis problem directly from data without constructing a model. Hsu et al. [24] provide a reach-avoid Q-learning algorithm with convergence guarantees for an arbitrarily tight conservative approximation of the reach-avoid set. Wang et al. [25] propose a falsification-based adversarial reinforcement learning algorithm for metric temporal logic specifications. Satisfying signal temporal logic specifications is studied by Verdier et al. [26] using counterexample-guided inductive synthesis on nonlinear systems, and using model-free reinforcement learning by Kalagarla et al. [27] to satisfy signal temporal logic specifications. A learning framework for synthesis of control-affine systems is provided by Sun et al. [28]. Watanabe et al. [29] study learning from demonstration while preventing the violation of safety under the learned policy. The recent papers [30,31] propose a data-driven approach to compute barrier certificates with correctness guarantees on satisfaction of safety specifications.

The research on data-driven constructions of abstract models is very limited. Legat et al. [32] provide an abstraction-based controller synthesis approach for hybrid systems by computing Lyapunov functions and Bellman-like Q-functions, and using a branch and bound algorithm to solve the optimal control problem. This differs from our approach where we want to satisfy temporal specifications instead of solving optimal control problems. Makdesi et al. [33] studied unknown monotone dynamical systems and sampled a set of trajectories generated by the system to find a minimal map overapproximating the dynamics of any system that produces these transitions. Consequently, they calculate an abstraction of the system related to this map and prove that an alternating bisimulation relation exists between them. In contrast to this work, our approach is not restricted to monotone systems and is applicable to any nonlinear dynamical system. Abstract models are also constructed for stochastic systems using sampled data.

Data-driven construction of abstract models for stochastic systems has also been studied recently. Badings et al. [34,35] consider constructing abstract models in the form of interval Markov decision processes (IMDPs) by computing probably approximately correct (PAC) bounds on the transition probabilities of the system. This makes the approach applicable for satisfying infinite-horizon specifications and providing confidence bounds on the (probabilistic) satisfaction of the specification. The work by Lavaei et al. [36] constructs finite MDPs using data for general nonlinear stochastic systems utilising the concept of stochastic bisimulation functions. The focus of these works is on stochastic systems, but our paper develops the results for non-probabilistic systems.

Since our results rely on knowing a possibly conservative bound on the system's Lipschitz constant, we review here the corresponding literature for finding such a bound. Lipschitz learning algorithms have been proposed to estimate a Lipschitz constant of a function under the assumption of knowing the function, e.g., for neural networks [37–39]. When the function is not known, available approaches [8,40,41] use and improve a traditional estimator by Strongin [42]. All these approaches provide proof of convergence to the true Lipschitz constant when the number of samples goes to infinity and can only provide an underapproximation of the system's Lipschitz constant for finite sample sizes. A recent result by Huang et al. [43] extends the Strongin's estimation method to handle bounded observational noise. It assumes having prior knowledge of an upper bound on the second-order partial derivatives of the function. The authors use least squares regression and provide a guarantee on the sample complexity of the approach for both

noiseless and noisy samples. Additionally, the authors obtain a theoretical minimum for the sample complexity, showing that their algorithm performs optimally without noise and near-optimally with noise. Under the assumption of knowing an upper bound over the second partial derivatives, the proposed method provides both asymptotic and finite sample guarantees that give closeness to the true Lipschitz constant with a certain confidence. In our approach, the underlying analytical model of the system is assumed to be unknown. The provided sample complexity result is based on the assumption of knowing the system's Lipschitz constant, which can be estimated using e.g., the approach of [8]. This is presented in Section 4.3. Our result still requires assuming that the estimation gives a correct bound. Alternatively, the recent approach of Huang et al. [43] can be used to estimate the Lipschitz constant with finite number of samples with a certain confidence under the assumption of knowing a bound on the second-order partial derivatives of the system trajectories.

The closest works to our problem formulation is the work by Devonport et al. [44] and the work of Xue et al. [45], where data-driven abstraction techniques are provided for satisfying finite-horizon specifications. Our results are more general and provide stronger guarantees in two main aspects. First, our constructed abstraction can be used for synthesising a controller against any linear temporal logic (LTL) specification and is not restricted to a fragment of LTL specifications. Our sample complexity result is independent of the horizon of the specification and does not limit using the approach on finite-horizon specifications. Second, the guarantee provided by Devonport et al. and by Xue et al. are based on PAC bounds, which means the constructed abstraction is always wrong on a small subset of the state space whose size can be made smaller at the cost of high computational efforts, and the approach will require an infinite number of samples if the size of this subset is set to zero. Our formulated guarantee ensures that the abstraction is valid on the *entire* state space with high confidence (i.e., confidence close to 1). The confidence is specified by $(1-\beta)$ in this paper and is interpreted from the frequentist view of probability: if we run our algorithm multiple times, we always get a correct abstraction except for a small number of times reflected in the confidence value. Having such a confidence value is essential in our approach since it relies on data gathered from the system. Smaller values of β gives higher confidence on getting a correct abstraction. This in turn increases the computational complexity of our approach since β appears directly in our sample complexity results.

In our approach, we formulate the synthesis problem as a robust convex program and approximate it with a scenario program. Such approximations have been studied for the past two decades. Calafiore and Campi [46] provide an approximately feasible solution for the associated chance-constrained program by solving a scenario program, and give a sample complexity result. Relaxing the convexity assumption is studied by Soudjani and Majumdar [47] by assuming additional properties of the underlying probability distributions. We will use the results by Esfahani et al. [48], where the optimality of the robust program is connected directly to the scenario program for performing data-driven verification and synthesis. Inspired by the works of Wood and Zhang [8], and Weng et al. [9], we will use extreme value theory to estimate the Lipschitz constant needed for the sample complexity results. Our results still require assuming that the estimation gives a correct bound.

2. Preliminaries and problem statement

2.1. Preliminaries

Notation. We denote the set of natural, real, positive real, and non-negative real numbers by $\mathbb{N}$, $\mathbb{R}$, $\mathbb{R}_{>0}$, and $\mathbb{R}_{\geq 0}$, respectively. The set of natural numbers including zero is denoted by $\mathbb{N}_{\geq 0}$. We use superscript $n > 0$ with these sets to denote the Cartesian product of n copies of these sets. The power set of a set A is denoted by 2^A and includes all the subsets of A. For any $x, y \in \mathbb{R}^n$ with $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$, and a relational symbol $\rhd \in \{\leq, <, =, >, \geq\}$, we write $x \rhd y$ if $x_i \rhd y_i$ for every $i \in \{1, 2, \dots, n\}$. A matrix $M \in \mathbb{R}^{n \times n}$ is said to be non-negative if all of its entries are non-negative. We use the operators $|\cdot|$ and $\|\cdot\|$ to denote the element-wise absolute value and the infinity norm, respectively. We use the notation $\Omega_\varepsilon(c) := \{x \in \mathbb{R}^n \mid \|x - c\| \leq \varepsilon\}$ to denote the ball with respect to infinity norm centred at $c \in \mathbb{R}^n$ with radius $\varepsilon \in \mathbb{R}^n_{>0}$. We consider a probability space $(\Omega, \mathcal{F}_\Omega, \mathbb{P}_\Omega)$, where Ω is the sample space, $\mathcal{F}_\Omega$ is a sigma-algebra on Ω comprising its subsets as events, and $\mathbb{P}_\Omega$ is a probability measure that assigns probabilities to events.

Control Systems. A continuous-time control system is a tuple $\Sigma = (X, x_{\text{in}}, U, W, f)$, where $X \subset \mathbb{R}^n$ is the state space, $x_{\text{in}} \in X$ is the initial state, $U \subset \mathbb{R}^m$ is the input space, and $W \subset \mathbb{R}^n$ is the disturbance space which is assumed to be a compact set containing the origin. The vector field $f : X \times U \to X$ is such that $f(\cdot, u)$ is locally Lipschitz for all $u \in U$. The evolution of the state of Σ is characterised by the differential equation

$$\dot{x}(t) = f(x(t), u(t)) + w(t), \tag{1}$$

where $w(t) \in W$ represents the additive disturbance.

We consider the class of input and disturbance signals $u : \mathbb{R}_{\geq 0} \to U$ and $w : \mathbb{R}_{\geq 0} \to W$ to be piecewise constant with respect to a *sampling time* $\tau > 0$, i.e., $u(t) = u(k\tau)$ and $w(t) = w(k\tau)$ for every $k\tau \leq t < (k+1)\tau$ and $k \in \mathbb{N}_{\geq 0}$.

Trajectories of Control Systems. In this paper, we consider control systems $\Sigma = (X, x_{\text{in}}, U, W, f)$ whose vector field f is not known, but we can observe their *time-sampled trajectories*. In order to define time-sampled trajectories, we need to first define *continuous-time trajectories* of control systems. Given a sampling time $\tau > 0$, an initial state $x_0 \in X$, a constant input $u \in U$, and a constant disturbance $w \in W$, define the continuous-time trajectory $\zeta_{x_0,u,w}$ of the system on the time interval $[0, \tau]$ as an absolutely continuous function $\zeta_{x_0,u,w} : [0, \tau] \to X$ such that $\zeta_{x_0,u,w}(0) = x_0$, and $\zeta_{x_0,u,w}$ satisfies the differential equation $\dot{\zeta}_{x_0,u,w}(t) = f(\zeta_{x_0,u,w}(t), u) + w$ for almost all

$t \in [0, \tau]$. The solution of (1) from x_0 for the constant control input u with $w(t) = 0$ for all $t \geq 0$ is called the *nominal trajectory* of the system. For a fixed τ, we define the operators

$$\varphi(x, u, w) := \zeta_{x,u,w}(\tau) \text{ and } \Phi(x, u) := \{\varphi(x, u, w) \mid w \in W\}$$

respectively for the trajectory at time τ and the set of such trajectories starting from x. A sequence $x_0, x_1, x_2, \ldots$ is a time-sampled trajectory of Σ if for each $i \geq 0$, we have $x_{i+1} \in \Phi(x_i, u_i)$ for some $u_i \in U$.

Remark 1. The effect of general disturbance signals $\{w : \mathbb{R}_{\geq 0} \to W'\}$ that are not necessarily piecewise constant, on the time-sampled trajectory $x_0, x_1, x_2, \ldots$ can be overapproximated by a set of piecewise constant disturbances $\{w : \mathbb{R}_{\geq 0} \to W\}$ such that $W' \subset W$. The derivation of W from W' depends on the sampling time τ and the continuity properties of the system.

We note that selecting a sampling time τ and considering time-sampled trajectories is essential for defining linear temporal specifications in their full generality, as defined next.

LTL Specifications. Our control tasks are defined using linear temporal logic (LTL). Here, we give a brief introduction to LTL. For detailed syntax and semantics of LTL, we refer to the book by Baier and Katoen [1] and references therein. We consider LTL specifications with syntax

$$\psi := \text{true} \mid p \mid \neg\psi \mid \psi_1 \wedge \psi_2 \mid \bigcirc\psi \mid \psi_1 \mathsf{U} \psi_2,$$

where $p \subset \mathbb{R}^n$ is an element of the set of atomic propositions AP. Let $\rho = x_0, x_1, \ldots$ be an infinite sequence of elements from $\mathbb{R}^n$ and denote $\rho_i = x_i, x_{i+1}, \ldots$ for any $i \in \mathbb{N}$. Then the satisfaction relation between ρ and a property ψ, expressed in LTL, is denoted by $\rho \vDash \psi$. We denote $\rho \vDash p$ if $x_0 \in p$. Furthermore, $\rho \vDash \neg\psi$ if $\rho \nvDash \psi$ and $\rho \vDash \psi_1 \wedge \psi_2$ if $\rho \vDash \psi_1$ and $\rho \vDash \psi_2$. For *next* operator, $\rho \vDash \bigcirc\psi$ holds if $\rho_1 \vDash \psi$. The temporal until operator $\rho \vDash \psi_1 \mathsf{U} \psi_2$ holds if $\exists i \in \mathbb{N} : \rho_i \vDash \psi_2$, and $\forall j \in \mathbb{N} : 0 \leq j < i, \rho_j \vDash \psi_1$. Disjunction ($\vee$) can be defined by $\rho \vDash \psi_1 \vee \psi_2 \Leftrightarrow \rho \vDash \neg(\neg\psi_1 \wedge \neg\psi_2)$. The operator $\rho \vDash \Diamond\psi$ is used to denote that the property will eventually happen at some point in the future. The operator $\rho \vDash \Box\psi$ signifies that ψ must always be true at all times in the future. We also define $\psi_1 \to \psi_2$ with $\neg\psi_1 \vee \psi_2$.

A few examples of LTL specifications in realistic applications are as follows. A robot visits two regions p_1 and p_2 infinitely often is denoted by $\psi = (\Box\Diamond p_1) \wedge (\Box\Diamond p_2)$. The frequency of the power network must always remains inside the range $p_1 = [49.5, 50.2]$ Hertz at all times and return back to $p_2 = [49.8, 50]$ if it goes down to $p_3 = [49.5, 49.8]$. This specification can be written as $\psi = \Box p_1 \wedge \Box(p_3 \to \Diamond p_2)$. The autonomous vehicle overtaking a parked vehicle while maintaining a minimum safe distance can be written as $\psi = p_1 \mathsf{U} p_2$, where p_1 indicates the safe region and p_2 the target location of the vehicle after overtaking.

Feedback Controller. A *feedback controller* for Σ is a function $C : X \to U$. We denote by $C \parallel \Sigma$ the feedback composition of Σ and C. The set of trajectories of the closed-loop system $C \parallel \Sigma$ consists of all finite trajectories $x_0, x_1, x_2, \ldots$ such that for all $i \in \mathbb{N}_{\geq 0}$, we have $x_{i+1} \in \Phi(x_i, C(x_i))$.

Finite Abstraction of Control Systems. Let $\Sigma = (X, x_{\text{in}}, U, W, f)$ be a control system, $\tau > 0$ be a fixed sampling time, and $\eta_x \in \mathbb{R}^n_{>0}$ and $\eta_u \in \mathbb{R}^m_{>0}$ denote the discretisation parameters for state and input spaces, respectively. Let $\widehat{X} \subset X$ be a finite set of points that are equally spaced with respect to the state space discretisation parameter η_x, and $\widehat{U}$ be a finite subset of equally spaced points the state input discretisation parameter η_u. A *finite-state abstraction* of Σ is denoted by a tuple $\widehat{\Sigma} = (\widehat{X}, \widehat{U}, \hat{f})$, where $\hat{x}'$ in $\hat{f}(\hat{x}, \hat{u})$ if there is a pair of states $x \in \Omega_{\eta_x}(\hat{x})$ and $x' \in \Omega_{\eta_x}(\hat{x}')$ such that $x' \in \Phi(x, \hat{u})$. Note that, the larger η_x is (where comparison is made dimension-wise), the smaller is the cardinality of $\widehat{X}$ resulting in a coarser abstraction. On the other hand, the smaller η_x is, the more precise the abstraction $\widehat{\Sigma}$ will be, increasing the chance of a successful controller synthesis (see, e.g., [3] for more details on this construction).

Feedback Refinement Relation. Let Σ be a control system and $\widehat{\Sigma}$ be its finite-state abstraction. A *feedback refinement relation* (FRR) from Σ to $\widehat{\Sigma}$ is a relation $Q \subseteq X \times \widehat{X}$ s.t. for all $x \in X$ there is some $\hat{x} \in \widehat{X}$ such that $Q(x, \hat{x})$ and for all $(x, \hat{x}) \in Q$, we have (i) $\widehat{U} \subseteq U$, and (ii) $u \in \widehat{U} \Rightarrow Q(f(x, u)) \subseteq \hat{f}(\hat{x}, u)$. We write $\Sigma \preceq_Q \widehat{\Sigma}$ if Q is an FRR from Σ to $\widehat{\Sigma}$.

Abstraction-based Controller Synthesis. Our synthesis objective is expressed as LTL specifications. The abstraction-based controller design (ABCD) [7] is a 3-step method to find a robust controller for the control system Σ. First, we compute a finite state abstraction $\widehat{\Sigma}$ s.t. $\Sigma \preceq_Q \widehat{\Sigma}$. Second, we synthesise an abstract controller of the form $\widehat{C} : \widehat{X} \to \widehat{U}$ for $\widehat{\Sigma}$ using methods from the reactive synthesis literature. Finally, we obtain the desired controller C as $C := \widehat{C} \circ Q$. It is known that this three step process produces a controller C such that $C \parallel \Sigma$ satisfies the specification [7]. For the details of the tool implementation using abstract models we refer to [49].

2.2. Problem statement

We study abstraction-based control design (ABCD) for systems with *unknown* dynamics using available data from the system such that a given specification is satisfied with high confidence on the closed-loop system.

Assumption 1. The vector field f of the control system $\Sigma = (X, x_{\text{in}}, U, W, f)$ in unknown, but sampled trajectories of the system can be obtained in the form of $S_N := \{(x_k, u_k, x'_k) \mid x'_k \in \Phi(x_k, u_k),\ k = 1, 2, \ldots, N\}$.

Problem 1 (*Data-Driven ABCD*)**. Inputs:** Control system $\Sigma = (X, x_{\text{in}}, U, W, f)$ with unknown vector field f, specification Ψ, sampled trajectories S_N, and confidence parameter $\beta \in (0, 1)$. **Outputs:** Abstract model $\widehat{\Sigma}$, abstract controller $\widehat{C}$, and refined controller C for Σ, such that $C \parallel \Sigma$ satisfies Ψ with confidence $(1 - \beta)$.

The first step of the ABCD is to compute a finite abstraction $\widehat{\Sigma}$ for Σ. Once such an abstraction is computed, synthesis of the controller $\widehat{C}$ and refining it to C follow the model-based ABCD scheme. Therefore, the main challenge is to provide a data-driven computation of the abstraction $\widehat{\Sigma}$ that is a true overapproximation of Σ with confidence $(1-\beta)$.

Problem 2 (*Data-Driven Abstraction*). **Inputs:** Control system $\Sigma = (X, x_{\text{in}}, U, W, f)$ with unknown vector field f, sampled trajectories $\mathcal{S}_N$, discretisation parameters η_x and η_u, and confidence parameter $\beta \in (0,1)$. **Outputs:** Finite model $\widehat{\Sigma}$ that is an abstraction of Σ with confidence $(1-\beta)$.

In this paper, we tackle Problem 2 by showing how to construct $\widehat{\Sigma}$ from sampled trajectories $\mathcal{S}_N$, and provide a lower bound on the data size N in order to ensure correctness of the abstraction with confidence $(1-\beta)$. The required theoretical tools are presented in the next section.

3. Robust convex programs

In this section, we describe *robust convex programs* (RCPs) and the data-driven approximation of their solution. In Sections 4 and 5, we show how such an approximation can be used for solving the data-driven abstraction in Problem 2.

Let $T \subset \mathbb{R}^q$ be a compact convex set for some $q \in \mathbb{N}$ and $c \in \mathbb{R}^q$ be a constant vector. Let $(\mathcal{D}, \mathscr{B}, \mathbb{P})$ be the probability space of the *uncertainty* and $g : T \times \mathcal{D} \to \mathbb{R}$ be a measurable function, which is convex in the first argument for each $d \in \mathcal{D}$, and bounded in the second argument for each $\theta \in T$. The RCP is defined as

$$\text{RCP:} \begin{cases} \min_\theta c^\top \theta \\ s.t.\ \theta \in T \text{ and } \ g(\theta, d) \le 0 \quad \forall d \in \mathcal{D}. \end{cases} \tag{2}$$

Computationally tractable approximations of the optimal solution of the RCP (2) can be obtained using *scenario convex programs* (SCPs) that only require gathering finitely many samples from the uncertainty space [50]. Let $(d_i)_{i=1}^N$ be N independent and identically distributed (i.i.d.) samples drawn according to the probability measure $\mathbb{P}$. The SCP corresponding to the RCP (2) strengthened with $\gamma \ge 0$ is defined as

$$SCP_\gamma : \begin{cases} \min_\theta c^\top \theta \\ s.t.\ \theta \in T, \text{ and } \ g(\theta, d_i) + \gamma \le 0 \ \forall i \in \{1, 2, \dots, N\}. \end{cases} \tag{3}$$

We denote the optimal solution of RCP (2) as θ^*_{RCP} and the optimal solution of SCP_γ (3) as θ^*_{SCP}. Note that θ^*_{RCP} is a single deterministic quantity but θ^*_{SCP} is a random quantity that depends on the i.i.d. samples $(d_i)_{i=1}^N$ drawn according to $\mathbb{P}$. The RCP (2) is a challenging optimisation problem since the cardinality of $\mathcal{D}$ is infinite and the optimisation has infinite number of constraints. In contrast, the SCP (3) is a convex optimisation with finite number of constraints for which efficient optimisation techniques are available [51]. The following theorem provides a sample complexity result for connecting the optimal solution of the SCP_γ to that of the RCP.

Theorem 1 (*[50]*). *Assume that the mapping $d \mapsto g(\theta, d)$ in (2) is Lipschitz continuous uniformly in $\theta \in T$ with Lipschitz constant L_d and let $h : [0,1] \to \mathbb{R}_{\ge 0}$ be a strictly increasing function such that*

$$\mathbb{P}(\Omega_\varepsilon(d)) \ge h(\varepsilon), \tag{4}$$

for every $d \in \mathcal{D}$ and $\varepsilon \in [0,1]$. Let θ^*_{RCP} be the optimal solution of the RCP (2) and θ^*_{SCP} the optimal solution of SCP_γ (3) with

$$\gamma = L_d h^{-1}(\varepsilon) \tag{5}$$

computed by taking N i.i.d. samples $(d_i)_{i=1}^N$ from $\mathbb{P}$. Then θ^*_{SCP} is a feasible solution for the RCP with confidence $(1-\beta)$ if the number of samples $N \ge N(\varepsilon, \beta)$, where

$$N(\varepsilon, \beta) := min \left\{ N \in \mathbb{N} \,\middle|\, \sum_{i=0}^{q-1} \binom{N}{i} \varepsilon^i (1-\varepsilon)^{N-i} \le \beta \right\}, \tag{6}$$

with q being the dimension of the decision vector $\theta \in T$.

4. Data-driven abstraction

In this section, we first discuss the steps required for model-based abstraction of control systems. We then show how this can be formulated as an RCP and present its associated SCP. Finally, we use the connection between the RCPs and SCPs in Theorem 1 to provide a lower bound for number of required samples to certify a desired confidence. The simplifying assumption used in this section is that samples from the *nominal trajectories* of the system Σ are also available in the form of $\{(x_k, u_k, x'_k) \mid x'_k = \varphi(x_k, u_k, 0),\ k = 1, 2, \dots, N\}$. We discuss in the next section how this assumption can be relaxed by modifying the inequality of a growth bound.

4.1. Growth bound for reachable sets

Consider a control system $\Sigma = (X, x_{\text{in}}, U, W, f)$ with the disturbance set $W = [-\bar{w}, \bar{w}]$ for some vector $\bar{w} \in \mathbb{R}^n_{\geq 0}$. Let η_x and η_u be discretisation parameters for the state and input spaces X and U used to construct $\widehat{X}$ and $\widehat{U}$ of sizes n_x and n_u, respectively. The first step of ABCD is to compute a finite abstraction $\widehat{\Sigma} = (\widehat{X}, \widehat{U}, \hat{f})$ using overapproximations of the reachable sets for every pair of abstract state and input. The reachable set for every pair $(\hat{x}, \hat{u}) \in \widehat{X} \times \widehat{U}$ is defined as

$$Reach(\hat{x}, \hat{u}) := \{x' \in \Phi(x, \hat{u}) \mid x \in \Omega_{\eta_x}(\hat{x})\}.$$

The set $Reach(\hat{x}, \hat{u})$ is usually overapproximated using a growth bound of the system dynamics [7].

Definition 4.1. For a control system Σ with abstract state and input spaces $\widehat{X}, \widehat{U}$, a function $\kappa : \mathbb{R}^n_{\geq 0} \times \widehat{X} \times \widehat{U} \to \mathbb{R}^n_{\geq 0}$ is called a *growth bound function* for Σ if it satisfies

$$|\varphi(x, \hat{u}, w) - \varphi(\hat{x}, \hat{u}, 0)| \leq \kappa(|x - \hat{x}|, \hat{x}, \hat{u}), \quad \forall \hat{x} \in \widehat{X},\ \forall \hat{u} \in \widehat{U},\ \forall x \in \Omega_{\eta_x}(\hat{x}),\ \forall w \in W. \tag{7}$$

Note that $\varphi(\hat{x}, \hat{u}, 0)$ is the nominal (disturbance-free) trajectory of the system. Using this definition, for every abstract state-input pair $(\hat{x}, \hat{u}) \in \widehat{X} \times \widehat{U}$, the reachable set $Reach(\hat{x}, \hat{u})$ is overapproximated with a ball centred at $z(\hat{x}, \hat{u}) := \varphi(\hat{x}, \hat{u}, 0)$ with radius $\lambda(\hat{x}, \hat{u}) := \kappa(\eta_x, \hat{x}, \hat{u})$.

When the system dynamics are known, it is shown by Reissig et al. [7] that a growth bound for the system can be computed as

$$\kappa(r, \hat{x}, \hat{u}) = e^{L(\hat{u})\tau} r + \int_0^\tau e^{L(\hat{u})s} \bar{w} ds, \tag{8}$$

for all $r \in \mathbb{R}^n_{\geq 0}$, $\hat{x} \in \widehat{X}$, and $\hat{u} \in \widehat{U}$, where $\bar{w}$ is the upper bound of the disturbance and $L : \widehat{U} \to \mathbb{R}^{n \times n}$ is a matrix such that the entries of $L(\hat{u})$ satisfy the following inequality for all $x \in X$:

$$L_{i,j}(\hat{u}) \geq \begin{cases} D_j f_i(x, \hat{u}) & i = j \\ |D_j f_i(x, \hat{u})| & i \neq j, \end{cases} \tag{9}$$

for all $i, j \in \{1, 2, \ldots, n\}$, where $f_i(x, u)$ is the ith element of the vector field $f(x, u)$ and $D_j f_i$ is its partial derivative with respect to the jth element of x.

4.2. SCP for the computation of growth bound

When the model of the system is unknown, a growth bound in (8) is not available since the matrix $L(\hat{u})$ defined using (9) is not computable. To tackle this bottleneck, we aim at computing a growth bound for the system that has the following parameterised form

$$\kappa_\theta(r, \hat{x}, \hat{u}) := \theta_1(\hat{x}, \hat{u}) r + \theta_2(\hat{x}, \hat{u}), \qquad \forall r \in \mathbb{R}^n_{\geq 0}, \hat{x} \in \widehat{X}, \hat{u} \in \widehat{U}, \tag{10}$$

where $\theta_1 \in \mathbb{R}^{n \times n}$ and $\theta_2 \in \mathbb{R}^n$. We denote by $\theta \in \mathbb{R}^{n^2+n}$ the concatenation of columns of θ_1 and θ_2.

Remark 2. The parameterised growth bound in (10) is linear with respect to r similar to (8), but is more general and less conservative by allowing θ_1, θ_2 to depend on $\hat{x}$ (i.e., they are defined locally for each abstract state).

Theorem 2. *The parameterised growth bound in (10) can be computed by solving the following robust convex program*

$$\begin{cases} \min_\theta c^\top \theta \\ s.t.\ 0 \leq \theta \leq \bar{\theta},\ \textit{and}\ \forall x \in \Omega_{\eta_x}(\hat{x}),\ \forall w \in W, \\ \quad |\varphi(x, \hat{u}, w) - \varphi(\hat{x}, \hat{u}, 0)| - \kappa_\theta(|x - \hat{x}|, \hat{x}, \hat{u}) \leq 0, \end{cases} \tag{11}$$

where $c = [1, 1, \ldots, 1] \in \mathbb{R}^{n^2+n}$ and $\bar{\theta}$ is a sufficiently large positive vector.

Proof. The inequality in (11) is a reformulation of (7) with a special choice of κ in (10), We show that the optimisation (11) is in fact a robust convex program. Let $\mathcal{D} = \Omega_{\eta_x}(\hat{x}) \times W$ be the uncertainty space and

$$g(\theta, x, w) := |\varphi(x, \hat{u}, w) - \varphi(\hat{x}, \hat{u}, 0)| - \kappa_\theta(|x - \hat{x}|, \hat{x}, \hat{u})$$

for all $x \in \Omega_{\eta_x}(\hat{x})$ and $w \in W$ and fixed $(\hat{x}, \hat{u}) \in \widehat{X} \times \widehat{U}$. We need to show that g is convex in θ for each $(x, w) \in \mathcal{D}$ and bounded in (x, w) for every $\theta \in [0, \bar{\theta}]$. The convexity holds due to the parameterisation of κ_θ in (10) being linear with respect to the optimisation variables in θ. The boundedness holds due to the set $\mathcal{D}$ being compact and trajectories of the system being continuous. □

To construct the SCP$_\gamma$ associated with the RCP (11), we fix $\hat{x} \in \widehat{X}$ and $\hat{u} \in \widehat{U}$, consider a uniform distribution on the space $\mathcal{D} = \Omega_{\eta_x}(\hat{x}) \times W$ and obtain N i.i.d. sample trajectories $S_N = \{(x_i, \hat{u}, x_i') \mid x_i' \in \Phi(x_i, \hat{u}), i = 1, 2, \ldots, N\}$. Note that every x_i' corresponds

to a random disturbance $w_i \in W$. The SCP$_\gamma$ is

$$\begin{cases} \min_\theta c^\top \theta \\ s.t.\ 0 \le \theta \le \bar{\theta} \text{ and } \forall i \in \{1, \dots, N\}, \\ \quad |x'_i - x'_{nom}| - \theta_1(\hat{x}, \hat{u})|x_i - \hat{x}| + \theta_2(\hat{x}, \hat{u}) + \gamma \le 0, \end{cases} \tag{12}$$

where $x'_{nom} := \varphi(\hat{x}, \hat{u}, 0)$ and $\gamma \in \mathbb{R}_{\ge 0}$.

Assumption 2. Let $L_\varphi(\hat{u})$ be an upper bound for the Lipschitz constant of the system trajectories $\varphi(x, \hat{u}, w)$ with respect to (x, w), i.e., for all $x, x' \in \Omega_{\eta_x}(\hat{x})$ and $w, w' \in W$,

$$\|\varphi(x, \hat{u}, w) - \varphi(x', \hat{u}, w')\| \le L_\varphi(\hat{u})\|(x, w) - (x', w')\|. \tag{13}$$

We assume that a possibly conservative correct upper bound $L_\varphi(\hat{u})$ is known.

Theorem 3. *Let $|\widehat{X}| = n_x$ and $|\widehat{U}| = n_u$. Under Assumption 2, for any $\hat{x} \in \widehat{X}$ constructed with discretisation size η_x, any $\hat{u} \in \widehat{U}$, and the disturbance set $W = [-\bar{w}, \bar{w}]$, the optimal solution of (12) gives a growth bound for the system Σ corresponding to $(\hat{x}, \hat{u})$ with confidence $(1 - \beta/(n_x n_u))$, when the number of samples $N \ge N(\varepsilon, \beta/(n_x n_u))$ and*

$$\gamma = 4L_\varphi(\hat{u}) \sqrt[2n]{\varepsilon \prod_{i=1}^{n} \eta_x(i) \prod_{i=1}^{n} \bar{w}(i)}, \tag{14}$$

where $\varepsilon \in [0, 1]$, and n is the dimension of the state space.

Proof. We apply Theorem 2 to the RCP (11) for fixed $\hat{x} \in \widehat{X}$ and $\hat{u} \in \widehat{U}$. Define

$$g(\theta, x, w) := \max\{|\varphi(x, \hat{u}, w) - \varphi(\hat{x}, \hat{u}, 0)| - \theta_1(\hat{x}, \hat{u})|x - \hat{x}| - \theta_2(\hat{x}, \hat{u})\}, \tag{15}$$

where the $\max\{\cdot\}$ is applied to the elements of its argument that belongs to $\mathbb{R}^n$. Since the distribution on $D = \Omega_{\eta_x}(\hat{x}) \times W$ is uniform, we choose

$$h(\varepsilon) = \mathbb{P}(\Omega_\varepsilon(d)) = \frac{(\varepsilon/2)^{2n}}{\prod_{i=1}^{n} \eta_x(i) \prod_{i=1}^{n} \bar{w}(i)}$$

to satisfy the inequality (4). Note that $h(\varepsilon)$ gives the probability of choosing a point within the $2n-$ball $\Omega_\varepsilon(d)$ uniformly at random. We use Eq. (5) as $\gamma = L_d h^{-1}(\varepsilon)$ to get the value of γ in (14). It only remains to show that $g(\theta, x, w)$ is Lipschitz continuous with constant $L_d = 2L_\varphi(\hat{u})$. Note that $L_\varphi(\hat{u})$ is the upper bound on the Lipschitz constant of $\varphi(x, \hat{u}, w)$ with respect to (x, w), and satisfies (13) for all $x, x' \in \Omega_{\eta_x}(\hat{x})$ and $w, w' \in W$. Since $\|\theta_1(\hat{x}, \hat{u})\|$ can be bounded by $L_\varphi(\hat{u})$, we get that

$$\begin{aligned} &\|g(\theta, x, w) - g(\theta, x', w')\| \\ &\le \|\varphi(x, \hat{u}, w) - \varphi(x', \hat{u}, w')\| + \|\theta_1(\hat{x}, \hat{u})\|\|x - x'\| \\ &\le L_\varphi(\hat{u})\|(x, w) - (x', w')\| + L_\varphi(\hat{u})\|x - x'\| \\ &\le 2L_\varphi(\hat{u})\|(x, w) - (x', w')\|, \end{aligned}$$

Therefore, $g(\theta, x, w)$ is Lipschitz continuous with constant $2L_\varphi(\hat{u})$. This completes the proof. □

Remark 3. The statement of Theorem 3 holds under Assumption 2 that requires knowing a correct (possibly conservative) upper bound $L_\varphi(\hat{u})$ on the Lipschitz constant of the system trajectories. To compensate for conservative values of $L_\varphi(\hat{u})$, a smaller value of ε is chosen, which will require taking higher number of samples N.

Remark 4. We provide an algorithm in the next subsection for estimating L_φ using sampled trajectories of the system. We emphasise that the approach of the next subsection gives only an "estimate" of L_φ. The results of Theorem 3 remains valid under the assumption that such estimation methods return a correct upper bound for L_φ.

Remark 5. We also remark that as the above proof shows, any correct upper bound L_φ gives $\theta_1 = L_\varphi \mathbf{1}_{n \times n}$, which can be used directly to construct the abstraction. This however gives conservative results, which will be demonstrated on a case study in Section 6.

Corollary 1. *Assume the SCP$_\gamma$ is solved under Assumption 2 for each state-input pair $(\hat{x}, \hat{u}) \in \widehat{X} \times \widehat{U}$ with confidence $(1 - \beta/(n_x n_u))$. Then the abstract model constructed using the obtained growth bounds is a valid abstract model for Σ with confidence at least $(1 - \beta)$.*

Proof. Denote the optimal solution of SCP$_\gamma$ in (12) by θ^*. According to Theorem 3, the ball centred at $z(\hat{x}, \hat{u}) := x'_{nom}$ with radius $\lambda(\hat{x}, \hat{u}) = \kappa_{\theta^*}(\eta_x, \hat{x}, \hat{u}) + \gamma$ is a valid overapproximation of the reachable set from the state-input pair $(\hat{x}, \hat{u})$ with confidence at least $1 - \beta/(n_x n_u)$. Denote by $\mathcal{G}_{\hat{x}\hat{u}}$ as the event that this ball is a valid overapproximation. Then $\mathbb{P}(\bar{\mathcal{G}}_{\hat{x}\hat{u}}) \le \beta/(n_x n_u)$ with $\bar{\mathcal{G}}_{\hat{x}\hat{u}}$ being the

complement of the event $\mathcal{G}_{\hat{x}\hat{u}}$. The constructed abstract model being a valid abstract model for Σ is the event $\bigcap_{(\hat{x},\hat{u})} \mathcal{G}_{\hat{x}\hat{u}}$. Using basic laws (e.g., De Morgan's law from set theory and Boole's inequality), we get

$$\mathbb{P}\Big(\bigcap_{\hat{x},\hat{u}} \mathcal{G}_{\hat{x}\hat{u}}\Big) = 1 - \mathbb{P}\Big(\bigcup_{\hat{x},\hat{u}} \bar{\mathcal{G}}_{\hat{x}\hat{u}}\Big) \geq 1 - \sum_{\hat{x},\hat{u}} \mathbb{P}(\bar{\mathcal{G}}_{\hat{x}\hat{u}}) \geq 1 - \sum_{\hat{x},\hat{u}} \beta/(n_x n_u) = 1 - n_x n_u \beta/(n_x n_u) = 1 - \beta.$$

Intuitively, since the number of pairs $(\hat{x},\hat{u})$ is $n_x n_u$, the chance of getting an invalid growth bound in "at least" one instance of SCP_γ is bounded by β. Therefore, we get a sound abstraction that truly overapproximates the behaviour of the system with confidence $(1-\beta)$. □

Remark 6. The parameter $\varepsilon \in [0,1]$ gives a trade off between the required number of samples and the level of conservativeness applied to the SCP. Smaller ε results in a larger number of sample trajectories, but reduces the value of γ in (14) (less conservative constraints in the SCP and higher chance of finding a feasible solution). In contrast, larger ε results in a smaller number of sample trajectories but increases the value of γ.

Remark 7. The quantity $2n$ used in (14) is in fact the dimension of the sample space $D = \Omega_{\eta_x}(\hat{x}) \times W$. If the system does not have any disturbance (i.e., the system can be modelled as an ODE having deterministic trajectories), the sample space will be $D = \Omega_{\eta_x}(\hat{x})$ and its dimension n can be used in (14): $\gamma = 4L_\varphi(\hat{u}) \sqrt[n]{\varepsilon \prod_{i=1}^{n} \eta_x(i)}$. This will substantially reduce the number of required sample trajectories. Similarly, if the disturbance does not affect some of the state equations, $2n$ can be replaced by $(n+q)$ where q is the dimension of the disturbance set considered as a non-zero measure set.

Algorithm 1 uses the result of Corollary 1 to provide an algorithmic solution for Problem 2. This algorithm receives a confidence parameters β, divides it by the cardinality of $\hat{X} \times \hat{U}$ (i.e., $n_x n_u$), computes a growth bound for each pair $(\hat{x},\hat{u}) \in \hat{X} \times \hat{U}$ using the SCP_γ in (12) with confidence $1 - \beta/(n_x n_u)$, and constructs the abstraction using these growth bounds.

Algorithm 1: Data-Driven Abstraction

Data: (X, U, W) of a control system Σ, confidence β, discretisation parameters η_x, η_u
Compute the finite state and input sets $\hat{X}$ and $\hat{U}$ using η_x, η_u;
Define n_x and n_u as cardinalities of $\hat{X}$ and $\hat{U}$;
Choose $\varepsilon \in [0,1]$;
Set $N = N(\varepsilon, \frac{\beta}{n_x n_u})$ using Eq. (6);
Compute γ using Eq. (14);
for $\hat{x} \in \hat{X}$ **do**
  for $\hat{u} \in \hat{U}$ **do**
    $\hat{f}(\hat{x},\hat{u}) = \emptyset$;
    Consider the uncertainty space $D = \Omega_{\eta_x}(\hat{x}) \times W$;
    Select N i.i.d sample trajectories using uniform distribution over D;
    Simulate the nominal trajectory $(\hat{x}, \hat{u}, x'_{nom})$;
    Solve the SCP_γ (12) to get the optimiser $\theta^*(\hat{x},\hat{u})$;
    $z \leftarrow x'_{nom}$;
    $\lambda \leftarrow \kappa_{\theta^*}(\eta_x, \hat{x}, \hat{u}) + \gamma$;
    Find all states $\hat{x}' \in \hat{X}$ for which $\Omega_{\eta_x}(\hat{x}') \cap \Omega_\lambda(z) \neq \emptyset$ and add them to $\hat{f}(\hat{x},\hat{u})$;
  end
end
Result: $\hat{\Sigma} = (\hat{X}, \hat{U}, \hat{f})$ as a finite abstraction of Σ with confidence $(1-\beta)$, $\theta^*(\hat{x},\hat{u})$ as a growth bound for $\hat{x} \in \hat{X}, \hat{u} \in \hat{U}$

The finite abstraction $\hat{\Sigma}$ constructed by Algorithm 1 is a valid abstraction for Σ with confidence $(1-\beta)$. This means any controller $\hat{C}$ synthesised on $\hat{\Sigma}$ and refined to a controller C for Σ will satisfy the desired specification with confidence $(1-\beta)$ on the closed loop system $\Sigma \parallel C$. In the next section, we extend our approach to make it suitable for abstraction refinement in case there is no controller $\hat{C}$ satisfying the specification due to the conservatism of the approach.

4.3. Lipschitz constant estimation

For estimating L_φ in (13), we estimate an upper bound for the fraction

$$\Delta(\hat{u}) := \frac{\|\varphi(x,\hat{u},w) - \varphi(x',\hat{u},w')\|}{\|(x,w) - (x',w')\|}$$

that holds for all $x, x' \in X$ and $w, w' \in W$. We follow the line of reasoning in [8,9] and use the extreme value theory for the estimation.

Let us fix a $\delta > 0$ and assign uniform distribution to the pairs (x, w) and (x', w') over the domain

$$\{x, x' \in X,\ w, w' \in W \text{ with } \|(x, w) - (x', w')\| \leq \delta\}. \tag{16}$$

Then $\Delta(\hat{u})$ is a random variable with an unknown cumulative distribution function (CDF). Based on the assumption of Lipschitz continuity of the system, the support of the distribution of $\Delta(\hat{u})$ is bounded from above, and we want to estimate an upper bound for its support. We take $\mathfrak{n}$ sample pairs (x, w) and (x', w'), and compute $\mathfrak{n}$ samples $\Delta_1, \Delta_2, \dots, \Delta_{\mathfrak{n}}$ for $\Delta(\hat{u})$. The CDF of $\max\{\Delta_1, \Delta_2, \dots, \Delta_{\mathfrak{n}}\}$ is called the limit distribution of $\Delta(\hat{u})$. Fisher–Tippett–Gnedenko theorem [52, Thm. 16] says that if the limit distribution exists, it can only be one of the three family of extreme value distributions — the Gumbel class, the Fréchet class, and the reverse Weibull class. These CDF's have the following forms:

$$\text{Gumbel class:} \quad G(s) = \exp\left[-\exp\left[\frac{s-a}{b}\right]\right],\ s \in \mathbb{R}$$

$$\text{Fréchet class:} \quad G(s) = \begin{cases} 0 & \text{if } s < a \\ \exp\left[-[\frac{s-a}{b}]^{-c}\right] & \text{if } s \leq a \end{cases}$$

$$\text{Reverse Weibull class:} \quad G(s) = \begin{cases} \exp\left[-[\frac{a-s}{b}]^{c}\right] & \text{if } s < a \\ 1 & \text{if } s \leq a \end{cases}$$

where $a \in \mathbb{R}, b > 0, c > 0$ are respectively the location, scale and shape parameters of the distributions.

Among the above three distributions, only the reverse Weibull class has a support bounded from above. Therefore, the limit distribution of $\Delta(\hat{u})$ will be from this class and the location parameter a is such an upper bound. As a result, we can estimate the location parameter of the limit distribution of $\Delta(\hat{u})$ to get an estimation of the Lipschitz constant.

The approach is summarised in Algorithm 2. The most inner loop computes samples of $\Delta(\hat{u})$. The middle loop computes samples of $\max\{\Delta_1, \dots, \Delta_{\mathfrak{n}}\}$. The outer loop estimates the Lipschitz constant for each $\hat{u}$ by fitting a reverse Weibull distribution.

Remark 8. The approach presented in this subsection can only be used for *estimating* the Lipschitz constant, which can then be enlarged by a correcting factor greater than one to account for the effect of estimation using finite number of samples. Note that this factor can be selected depending on the system under study by fitting the Reverse Weibull distribution to datasets of varying size and observing its convergence behaviour. Our results are valid under the assumption that the (corrected) estimation provided by Algorithm 2 gives a valid upper bound on the Lipschitz constant of the system.

Algorithm 2: Lipschitz Constant Estimation

Data: (X, U, W) of a control system Σ, abstract input space $\widehat{U}$
Select number of samples $\mathfrak{n}$ and $\mathfrak{m}$ for the estimation
Select $\delta > 0$
for $\hat{u} \in \widehat{U}$ **do**
  for $j = 1 : \mathfrak{m}$ **do**
    for $i = 1 : \mathfrak{n}$ **do**
      Sample pairs $(x, w), (x', w')$ uniformly from the domain in (16)
      Run Σ to get trajectories $\varphi(x, \hat{u}, w)$ and $\varphi(x', \hat{u}, w')$
      Compute $\Delta_i := \frac{\|\varphi(x,\hat{u},w) - \varphi(x',\hat{u},w')\|}{\|(x,w)-(x',w')\|}$
    end
    $\Gamma_j := \max\{\Delta_1, \dots, \Delta_{\mathfrak{n}}\}$
  end
  Fit a reverse Weibull distribution to the sample set $\{\Gamma_1, \Gamma_2, \dots, \Gamma_{\mathfrak{m}}\}$
  $L_\varphi(\hat{u})$ is the location parameter of the fitted distribution
end
Result: Estimated value of $L_\varphi(\hat{u})$ for all $\hat{u} \in \widehat{U}$

Assumption 2 on having a possibly conservative bound on the Lipschitz constant of the system or correctness of the estimated bound from Algorithm 2 can be replaced by knowing an upper bound on the second-order partial derivatives of the system trajectories as stated formally in the next theorem.

Theorem 4. *Assume a bound on the second-order partial derivatives of the system trajectories is available. For any $\varepsilon_{\mathsf{e}} > 0$ and $\beta_{\mathsf{e}} \in (0, 1)$, there is an estimation algorithm with a sample size N_{e} that gives $\hat{L}_\varphi$ as an estimation of L_φ such that*

$$\mathbb{P}\left(|\hat{L}_\varphi - L_\varphi| \leq \varepsilon_{\mathsf{e}}\right) \geq 1 - \beta_{\mathsf{e}}.$$

Using $(\hat{L}_\varphi + \varepsilon_{\mathsf{e}})$ in Algorithm 1 as the bound on the Lipschitz constant generates a finite abstraction $\widehat{\Sigma}$ of Σ that is correct with confidence $(1 - \beta - \beta_{\mathsf{e}})$.

Proof. The first part follows the recent result [43, Thm. 3.10]. For the second part, define the two events

$$\mathcal{A}_e := \{L_\varphi \le \hat{L}_\varphi + \varepsilon_e\}, \quad \mathcal{B}_e := \{\widehat{\Sigma} \text{ is a correct abstraction of } \Sigma\}.$$

Then, $\mathbb{P}(\mathcal{A}_e) \ge 1-\beta_e$. Using the first part of the theorem, and also using the fact that $\mathbb{P}(\mathcal{B}_e \mid \hat{L}_\varphi + \varepsilon_e \ge L_\varphi) \ge 1-\beta$ using Corollary 1 applied to Algorithm 1 with constant $(\hat{L}_\varphi + \varepsilon_e)$. The law of total probability gives

$$\mathbb{P}(\mathcal{B}_e) = \mathbb{P}(\mathcal{B}_e \mid \mathcal{A}_e)\mathbb{P}(\mathcal{A}_e) + \mathbb{P}(\mathcal{B}_e \mid \bar{\mathcal{A}}_e)\mathbb{P}(\bar{\mathcal{A}}_e) \ge (1-\beta)(1-\beta_e) \ge 1-\beta-\beta_e. \quad \square$$

5. Synthesis via abstraction refinement

The data-driven synthesis discussed in Section 4 inherits the soundness property from the ABCD approach: they both work with overapproximations of the dynamics and may not return a controller even if one may exists. Therefore, there is a need for refining the abstraction in order to check for controllers using less conservative abstractions. While the method of Section 4 is good for a given fixed discretisation parameter η_x, it is not suitable for reducing η_x, which requires re-computing all local parameters of growth bounds $\theta_1(\hat{x},\hat{u}), \theta_2(\hat{x},\hat{u})$. Another shortcoming of the method is related to the data collection: the nominal trajectories of the system should be available and are used in the constraints of the SCP. In this section, we discuss an extension of the approach of Section 4, in order to

- enable reducing η_x without the need for re-computing a growth bound, and
- relax the assumption of having access to the nominal trajectories of the system.

Let us define a modified growth bound as a function $\kappa^e : \mathbb{R}^n_{\ge 0} \times \widehat{X} \times \widehat{U} \to \mathbb{R}^n_{\ge 0}$ that is strictly increasing in its first argument and satisfies

$$|\varphi(x_1,\hat{u},w_1) - \varphi(x_2,\hat{u},w_2)| \le \kappa^e(|x_1-x_2|,\hat{x},\hat{u}), \qquad \forall \hat{x}\in\widehat{X},\ \forall \hat{u}\in\widehat{U},\ \forall x_1,x_2 \in \Omega_{\eta_x}(\hat{x}),\ \forall w_1,w_2 \in W. \tag{17}$$

This definition is more conservative than (7) in comparing trajectories under two arbitrary disturbances, and we always have that κ^e satisfies (7). Using this new definition, for every pair of abstract state and input $(\hat{x},\hat{u})$, the corresponding overapproximation of the reach set can be computed as a ball centred at *any* $z(\hat{x},\hat{u}) \in \Phi(\hat{x},\hat{u})$ with radius $\lambda(\hat{x},\hat{u}) = \kappa^e(\eta_x,\hat{x},\hat{u})$.

we choose a parametrisation for κ^e similar to (10), i.e.,

$$\kappa^e_\theta(r,\hat{x},\hat{u}) = \theta_1(\hat{x},\hat{u})r + \theta_2(\hat{x},\hat{u}), \tag{18}$$

where $r \in \mathbb{R}_{\ge 0}$, $\theta_1 \in \mathbb{R}^{n\times n}$, $\theta_2 \in \mathbb{R}^n$, and $\theta \in \mathbb{R}^{n^2+n}$ is constructed by concatenating columns of θ_1 and θ_2. The SCP associated with this growth bound is constructed by considering a uniform distribution over $\Omega_{\eta_x}(\hat{x}) \times W$ and obtain $2N$ i.i.d. sample trajectories $\mathcal{S}_{2N} = \{(x_i,\hat{u}_i,x_i') \mid x_i' \in \Phi(x_i,\hat{u}), i = 1,2,\dots,2N\}$ so that every x_i' corresponds to a random disturbance $w_i \in W$. The modified SCP_γ is defined as

$$\begin{cases} \min c^\top \theta \\ s.t.\ 0 \le \theta \le \bar{\theta} \text{ and } \forall i \in \{1,\dots,N\} \\ |x'_{2i-1} - x'_{2i}| - \theta_1(\hat{x},\hat{u})|x_{2i-1} - x_{2i}| - \theta_2(\hat{x},\hat{u}) + \gamma \le 0 \end{cases}$$

where $c = [1,1,\dots,1] \in \mathbb{R}^{n^2+n}$ is a constant vector, $\bar{\theta} \in \mathbb{R}^{n^2+n}_{>0}$ is sufficiently large, and $\gamma \ge 0$.

Theorem 5. *For any $\hat{x} \in \widehat{X}$ constructed with the discretisation size η_x, any $\hat{u} \in \widehat{U}$, and the disturbance set $W = [-\bar{w},\bar{w}]$, the optimal solution of (19) gives a growth bound for the system Σ corresponding to $(\hat{x},\hat{u})$ that satisfies (17) with confidence $(1-\beta)$, when the number of samples $2N \ge N(\varepsilon,\beta)$ and*

$$\gamma = 8L_\varphi \sqrt[4n]{\varepsilon \left[\prod_{i=1}^n \eta_x(i) \prod_{i=1}^n \bar{w}(i)\right]^2}, \tag{19}$$

where $\varepsilon \in [0,1]$, n is the dimension of the state space, and $L_\varphi(\hat{u})$ is an upper bound on the Lipschitz constant of the system trajectories $\varphi(x,\hat{u},w)$ with respect to (x,w).

Proof. The proof of this theorem is similar to that of Theorem 3. Define

$$g(\theta,x_1,w_1,x_2,w_2) := \max\{|\varphi(x_1,\hat{u},w_1) - \varphi(x_2,\hat{u},w_2)| - \theta_1(\hat{x},\hat{u})|x_1-x_2| - \theta_2(\hat{x},\hat{u})\}.$$

To satisfy the inequality (4), we can choose

$$h(\varepsilon) = \mathbb{P}(\Omega_\varepsilon(d)) = \frac{(\varepsilon/2)^{4n}}{[\prod_{i=1}^n \eta_x(i) \prod_{i=1}^n \bar{w}(i)]^2},$$

since the distribution on $(\Omega_{\eta_x}(\hat{x}) \times W)^2$ is uniform. Using Eq. (5), we have $\gamma = L_d h^{-1}(\varepsilon)$. In order to prove that γ takes the value in (19), we must show that g is Lipschitz continuous with constant $L_d = 4L_\varphi(\hat{u})$. Bounding $\|\theta_1(\hat{x},\hat{u})\|$ by L_φ, for all (x_1,w_1,x_2,w_2) and (x_1',w_1',x_2',w_2') we have

$$\|g(\theta,x_1,w_1,x_2,w_2) - g(\theta,x_1',w_1',x_2',w_2')\|$$

Table 1
Results for the DC-DC boost converter.

Case-study	Dimension		Disturbance	Fixed discretisation		
	X	U	W	N	Time (min)	$\lvert\mathcal{V}\rvert$
DC-DC boost converter	2	1	$\{0\}$	1,807	22.2	37,783
			$[-0.01, 0.01]$	2,285	30.6	37,414

$$
\begin{aligned}
&\leq \|\varphi(x_1, \hat{u}, w_1) - \varphi(x_1', \hat{u}, w_1')\| \\
&+ \|\varphi(x_2, \hat{u}, w_2) - \varphi(x_2', \hat{u}, w_2')\| \\
&+ \|\theta_1(\hat{x}, \hat{u})\|(\|x_1 - x_1'\| + \|x_2 - x_2'\|) \\
&\leq 4L_\varphi(\hat{u})\|(x_1, w_1, x_2, w_2) - (x_1', w_1', x_2', w_2')\|.
\end{aligned}
$$

Therefore, g is Lipschitz continuous with constant $4L_\varphi(\hat{u})$. This completes the proof. □

A statement similar to Corollary 1 holds for a growth bound computed using (19).

6. Experimental evaluation

To demonstrate our approach, we apply it to a DC-DC boost converter and a path planning problem. These case studies are taken from Rungger and Zamani [49] and Girard et al. [53] and will be used as black-box models to generate sample trajectories. We also introduce a case study from power systems based on the work of Ma and Fan [54], that is implemented in the Power System Toolbox (PST) [55]. We will use trajectories from the black-box reduced model of the 30 state power system model. We apply our approach to construct finite abstractions of these systems and employ SCOTS [49] to design controllers. Our algorithms are implemented in C++ on a 64-bit Linux cluster machine with two Intel Xeon E5 v2 CPUs, 1866 MHz, and 50 GB RAM.

6.1. DC–DC boost converter

The objective in the DC-DC boost converter problem is to design a controller to enforce a reach and stay specification. The DC-DC boost converter can be modelled as a two dimensional linear switching system with two functional modes. The state vector of the system at time $t \in \mathbb{R}_{\geq 0}$ is $x(t) = (i_l(t), v_c(t))$, where i_l is the inductor current and v_c is the capacitor voltage. The system's evolution can be controlled by selecting the appropriate mode $u(t) \in \{1, 2\}$ at every time $t \in \mathbb{R}_{\geq 0}$. The system's dynamics under the two modes can be represented as $\dot{x} = A_{u(t)}x(t) + b + cw(t)$, $u \in \{1, 2\}$, with matrices A_1, A_2, b, c as reported in [53]. The state and input spaces are $X = [0.65, 1.65] \times [4.95, 5.95]$ and $U = [1, 2]$. The initial state is $(i_{l_0}(t), v_{c_0}(t)) = (0.7, 5.4)$ and the target set is $[1.1, 1.6] \times [5.4, 5.9]$. The target set is shown in red colour in Fig. 2. Our implementation results are reported in Table 1 for the system without disturbance ($\bar{w} = (0, 0)$) and with disturbance bound $\bar{w} = (0.01, 0)$. These results are obtained with discretisation parameters $\eta_x = (0.005, 0.005)$ and $\eta_u = 1$, confidence parameter $\beta = 0.01$, $\varepsilon = 0.01$ and estimation for $L_\varphi = 0.9935$. The resulted finite abstraction has cardinalities $n_x = 40{,}000$ and $n_u = 2$. The required number of sample trajectories, N, for each $(\hat{x}, \hat{u}) \in \hat{X} \times \hat{U}$ is computed using Eq. (6). Runtimes and the resulting winning region sizes, $|\mathcal{V}|$, for the DC-DC boost converter are given in Table 1.

We have used Algorithm 1 to compute the finite-state abstraction by collecting sample trajectories of the system. Subsequently, SCOTS is used for designing the controller. The performance of the controller is shown in Fig. 2 for the system without disturbance in the left and with disturbance in the right. These plots show one sample closed-loop trajectory of the system under the controllers designed by our data-driven ABCD approach. In both cases, without and with disturbance, it can be noticed from Fig. 2 that our approach has been successful in finding controllers satisfying the given reach and stay specification, despite the dynamics being unknown.

6.2. Path planning problem with partition refinement

We consider a path planning problem for a vehicle that is modelled as

$$
\begin{aligned}
\dot{x} &= v\cos(\alpha + \theta)/\cos(\alpha) + w \\
\dot{y} &= v\sin(\alpha + \theta)/\cos(\alpha) \\
\dot{\theta} &= v\tan(\omega),
\end{aligned}
\tag{20}
$$

where the state variables x, y, θ represent the position of the vehicle in the 2-dimensional space and the orientation of the vehicle, respectively. Inputs are (v, ω), the disturbance is w, and $\alpha := \arctan(\tan(\omega)/2)$. The state and input spaces are $X = [0, 10] \times [0, 10] \times [-\pi - 0.4, \pi + 0.4]$ and $U = [-1, 1]^2$, respectively. The goal is to find a controller to steer the vehicle from the initial state $(x_0, y_0, \theta_0) = (0, 1.2, 0)$ to the target set $(x, y) \in [9, 9.51] \times [0, 0.51]$ while avoiding the obstacles. These obstacles are shown in blue colour in Fig. 3.

We computed growth bounds with a coarse discretisation $\eta_x = (1.6, 1.6, 1.6)$ and reduced them iteratively with the factor of two. The algorithm successfully finds a controller for the system after five iterations. The implementation results are reported in Table 2. These results are obtained with $\eta_u = (0.3, 0.3)$, the confidence parameter $\beta = 0.01$, $\varepsilon = 0.01$ and estimated constant $L_\varphi = 1.46$. The

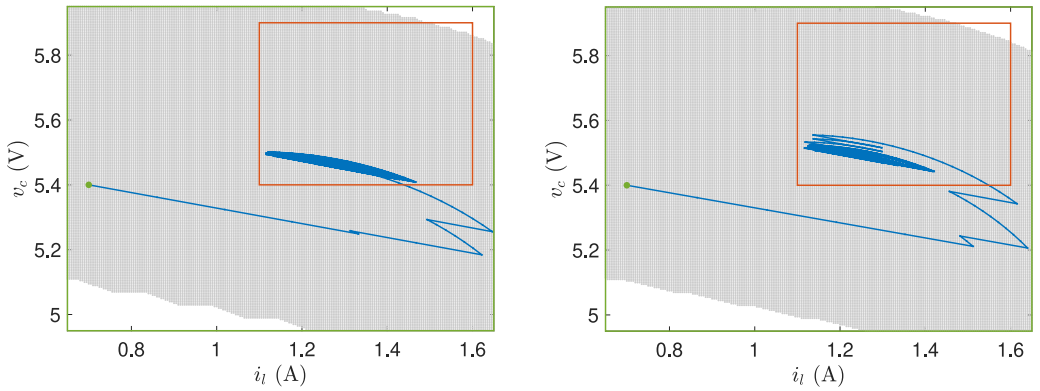


Fig. 2. The closed-loop trajectory of the DC-DC boost converter under the controller designed by our data-driven abstraction approach, the left with $\bar{w} = (0, 0)$ and the right with $\bar{w} = (0.01, 0)$. The rectangle in red colour represents the target region and the area in grey shows the winning region of the controller. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2
Results for the path planning case study.

Case-study	Dimension		Disturbance	Abstraction refinement		
	X	U	$\bar{w}$	N	Time (min)	$\lvert\mathcal{V}\rvert$
Path planning	3	2	$(0, 0, 0)$	3,127	225	405,493
			$(0.01, 0, 0)$	4,277	513	447,212

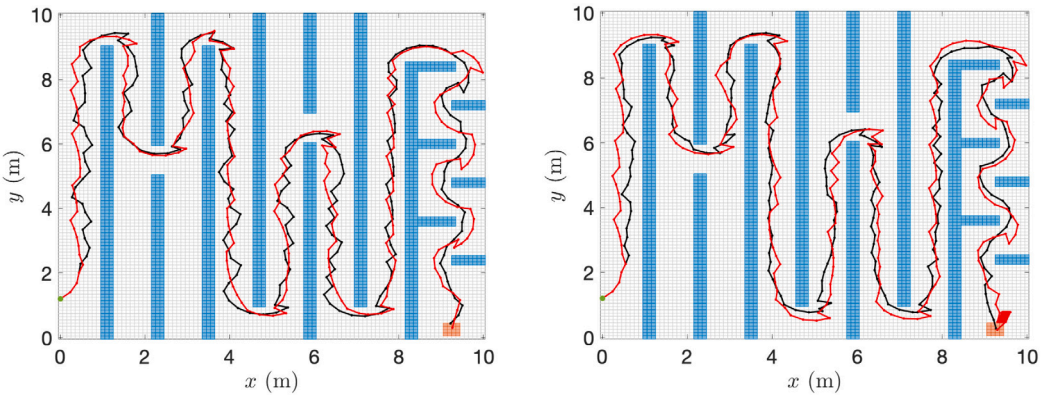


Fig. 3. Comparison between the closed-loop trajectories of the system (20) without disturbance in the left and with disturbance bound $\bar{w} = (0.01, 0, 0)$ in the right under the controllers designed by our data-driven abstraction refinement approach (black) and by the model-based approach of SCOTS (red). Blue blocks represent the obstacles, the green dot represents the initial state, and the orange rectangle shows the target region. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

resulted abstraction has cardinalities $n_x = 88{,}500$ and $n_u = 24$. For the case of disturbance-free model we set $\bar{w} = (0, 0, 0)$, and for the case of dynamics with disturbance, we set $\bar{w} = (0.01, 0, 0)$. The required number of sample trajectories for each $(\hat{x}, \hat{u})$ is computed using Eq. (6) and marked with N in the table. Finally, runtimes and size of the winning regions $|\mathcal{V}|$ are reported.

We have used the synthesis method based on abstraction refinement presented in Section 5, to construct the finite-state abstraction by collecting sample trajectories of the system. We used SCOTS to design the controller fulfilling the given specification. The performance of the controller is shown in Fig. 3 for the system without and with the disturbance, respectively. These figures compare the closed-loop trajectories of the system under the controllers designed by our data-driven abstraction refinement algorithm approach (black) and by the model-based approach of SCOTS (red). Our data-driven approach successfully finds a controller for the system that satisfies the specification without the need for knowing the dynamics of the system.

6.3. Three area three machine power system

We consider a three area three machine (3A3M) power system adapted from Ma and Fan [54], and shown in Fig. 4. The system consists of three buses, which are each connected to a power source (generator) and a load. At bus 1 we consider a load which is

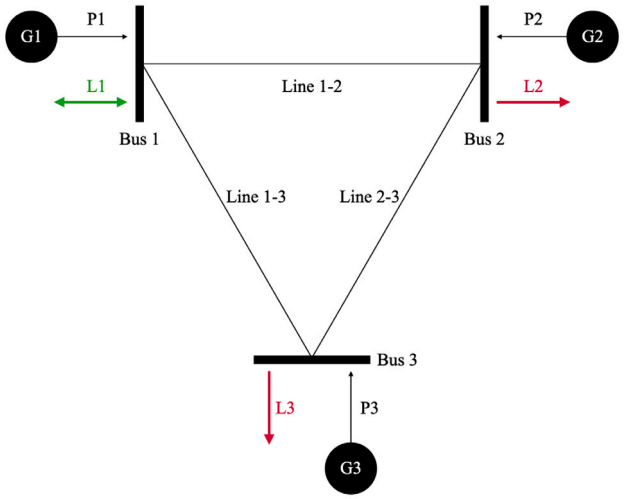


Fig. 4. 3A3M power system with generators (G) and loads (L). L1 represents a bidirectional load such as Electric Vehicles or Energy Storage Systems.

Table 3
Results for the 3A3M power system.

Control approach	Dimension		Disturbance			
	X	U	$\bar{w}$	N	Time (min)	$\lvert\mathcal{V}\rvert$
Fixed Discretisation	3	1	(0.2, 0.3)	3,290	5,253	230,760
Adaptive Refinement			(0.2, 0.3)	4,460	50.25	314,802

bidirectional, meaning it can both draw power and inject power into the system. The loads at buses 2 and 3 can only draw power from the system; when these loads increase, more power will be drawn from the system, causing an imbalance between generation and consumption which may result in a reduction of the network frequency. The nominal frequency of the network is set to 60 Hz.

We consider a worst-case scenario when a sudden increase occurs in the loads at buses 2 and 3 by 0.2 and 0.3 per unit (p.u.), respectively. The control task is for the load at bus 1 to balance the load increase at buses 2 and 3 by either reducing its load or injecting power into the network. The simulation is run using PST on a 30 state model of this power system. Balanced realisation of the system reduces its dynamics to three states. To compute the data-driven finite abstraction, sample trajectories are gathered using a black-box approach of the reduced system representation for the original model. The dynamics of the reduced system are given by

$$\begin{aligned} \dot{x} &= Ax + Bu + Ew \\ y &= Cx, \end{aligned} \tag{21}$$

where

$$A = \begin{bmatrix} 0.00027563 & 0 & 0 \\ 0 & -0.3951 & 0.687 \\ 0 & -0.6869 & -0.016 \end{bmatrix}, B = \begin{bmatrix} 0.00031166 \\ 0.1359 \\ 0.0230 \end{bmatrix}, E = \begin{bmatrix} 0.00033103 & 0.00031244 \\ 0.1309 & 0.1308 \\ 0.0250 & 0.0233 \end{bmatrix}, C = \begin{bmatrix} -0.0115 & -0.2296 & 0.0412 \end{bmatrix}. \tag{22}$$

The state and input spaces are $X = [-0.02, 0.02] \times [-0.05, 0.05] \times [-0.12, 0.12]$ and $U = [0, 0.5]$. Further, we set $W = [-0.2, 0.2] \times [-0.3, 0.3]$, $\eta_u = 0.025$, $\tau = 0.4$, $\eta_x = (0.0015, 0.0015, 0.0015)$, $\beta = 0.01$ and $\varepsilon = 0.01$. The resulted abstraction has $n_x = 228,480$ and $n_u = 20$. The estimated Lipschitz constant is $L_\varphi = 1.5715$. The target set is given by $-0.008 < y < 0.008$ and the avoid set is given by $y < -0.015$. Multiplying by the nominal frequency to get the specification in Hertz, the target region is $[59.52, 60.48]$ and the avoid region is $(-\infty, 59.1)$. Fig. 5 shows that the specification is violated when no control is applied.

We apply the data-driven approaches of Section 4 (fixed discretisation) and Section 5 (abstraction refinement). Both controllers are synthesised with disturbance $W = [-0.2, 0.2] \times [-0.3, 0.3]$. A comparison of the two control approaches is shown in Table 3. The required number of sample trajectories for each $(\hat{x}, \hat{u})$ is computed using Eq. (6) and marked with N in the table. The abstraction refinement starts with $\eta_x = 0.012$ and refines the discretisation iteratively with a factor of two. The algorithm successfully finds a controller after five iterations. The runtimes and the resulting winning region sizes $|\mathcal{V}|$ are also given in Table 3. The abstraction refinement synthesises the controller a factor of 100 times faster than the fixed discretisation by iteratively decreasing the value of η_x.

The data-driven control approach with fixed discretisation is simulated in PST and is reported in Figs. 6 and 7. The controlled system successfully keeps the frequencies of the three areas outside of the avoid set (i.e., always above 59.1 Hz) and bring them back

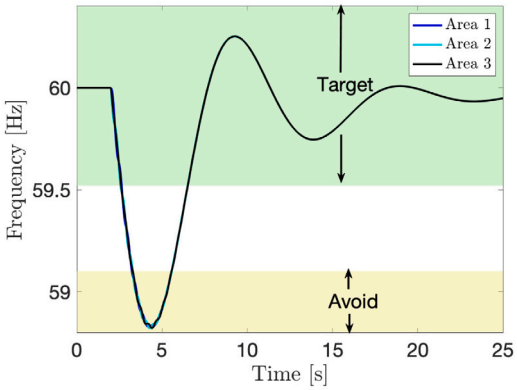


Fig. 5. 3A3M power system frequency without applying any control input. The frequency falls below 59.1 Hz thus violates the specification.

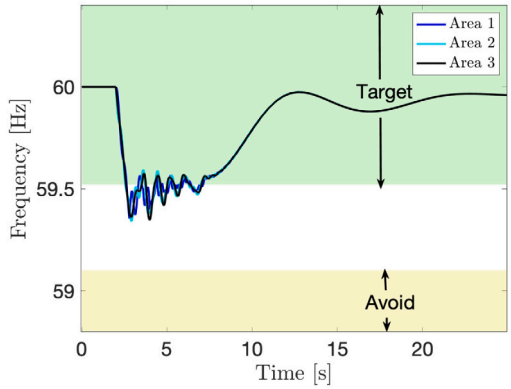


Fig. 6. 3A3M power system frequencies for the three areas, with the frequency of an area is measured at the corresponding bus in that area. The control synthesised by the fixed discretisation approach successfully keeps the frequencies of the three areas outside of the avoid set. The frequencies leave the target set for around 4.4 s before staying in the target set.

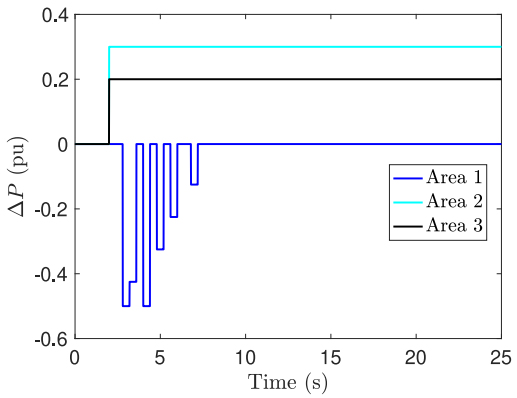


Fig. 7. 3A3M power system load changes for the three areas. Loads at buses 2 and 3 increase by 0.3 and 0.2 p.u., respectively. Load at bus 1 is used to control the frequency using our data-driven approach with fixed discretisation.

to the target set (i.e., above 59.52 Hz). Fig. 7 shows the load changes in the system. Load at bus 1 is able to maintain the frequencies of the three areas above the avoid region and facilitate the system returning to the target set for the maximum disturbances applied at buses 2 and 3. Figs. 8 and 9 show the results of simulating the system in PST with the control obtained from the abstraction refinement approach. The controlled system has the same performance in satisfying the specification.

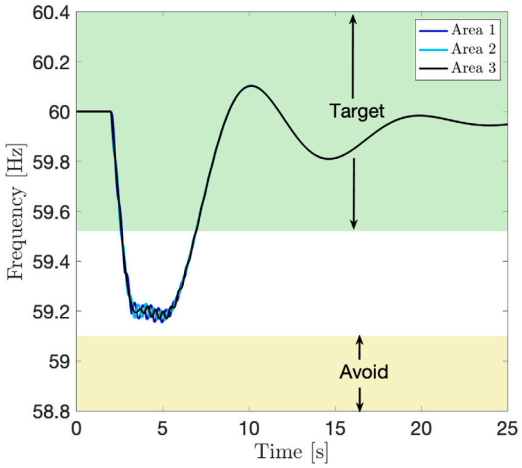


Fig. 8. 3A3M power system frequencies for the three areas, with the frequency of an area is measured at the corresponding bus in that area. The control synthesised by the abstraction refinement approach successfully satisfies the specification. The frequencies leave the target set for around 4.2 s before staying in the target set.

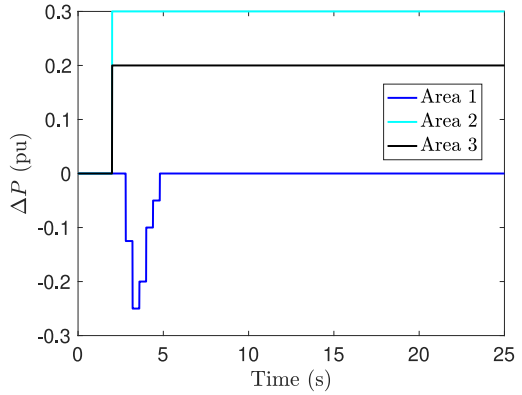


Fig. 9. 3A3M power system load changes for the three areas. Loads at buses 2 and 3 increase by 0.3 and 0.2 p.u., respectively. Load at bus 1 is used to control the frequency using our data-driven approach with abstraction refinement.

6.4. Comparison with PAC learning

In this subsection, we compare our approach with the results provided by Xue et al. [45] that is based on probably approximately correct (PAC) bounds on the 3A3M power system case study. The abstraction approach of Xue et al. has no bias term γ and requires the number of samples

$$N \geq \frac{2}{\nu}(ln\frac{1}{\beta} + q), \tag{23}$$

where $\beta \in (0,1)$ is the confidence parameter, $\nu \in (0,1)$ is the error threshold, and $q \in \mathbb{N}$ is the cardinality of the parameter vector θ. The error threshold allows the constructed abstraction to hold for the entire state space except a subset measured by parameter $\nu > 0$. Note that setting $\nu = 0$ in (23) results in requiring an infinite dataset, which is not practical. Our approach provides an abstraction that is correct on the entire state space, i.e., $\nu = 0$, with a finite sample size.

We implement our data-driven robust scenario approach (RSA), the PAC approach [45] with parameters $\beta = 0.01$ and $\nu = 0.01$, and the model-based approach [7]. Table 4 compares the winning domain of the controllers by reporting the intersections ($\cap$) and set differences (row $\setminus$ column). It can be seen that the winning domain obtained by our RSA method is a subset of the ones computed by PAC and the model-based approaches. This shows that our approach is more conservative than the model-based approach but correctly finds a subset of the winning domain. In contrast, the PAC approach gives a winning domain that includes states not identified as winning by the model-based approach. It includes 1079 states outside of the winning domain obtained by the model-based approach. Due to the nature of the PAC learning, some of these states are incorrectly identified as winning. The main reason is that the PAC method may miss capturing some of the transitions and does not always generate an overapproximation of the system behaviours. Among these 1079 states, a counter-example can be found, demonstrating a lack of guarantee provided by the PAC

Table 4
Comparing the winning domain of controllers obtained from our RSA method, PAC method of [45], and the model-based approach of [7]. The pairwise comparison is made by computing the intersections (∩) and set differences (row \ column). The results are reported both in cardinalities and percentages.

Winning domain	RSA		PAC		Model-based	
	∩	\	∩	\	∩	\
RSA	230,760	0	230,760	0	230,760	0
%	100.00%	0.00%	100.00%	0.00%	100.00%	0.00%
PAC	230,760	15,664	246,424	0	245,345	1,079
%	93.64%	6.36%	100.00%	0.00%	99.56%	0.44%
Model-based	230,760	22,216	245,345	7,631	252,976	0
%	91.22%	8.78%	96.98%	3.02%	100.00%	0.00%

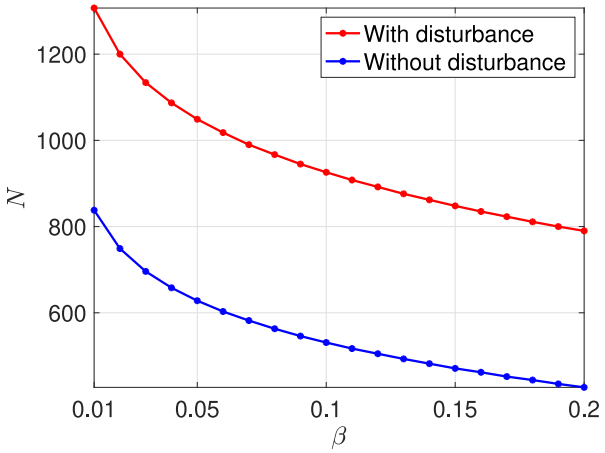


Fig. 10. Required number of samples for our approach as a function of β for a fixed $\epsilon = 0.01$.

method. At state $(0.0187, 0.0262, -0.1163)$ the PAC controller calculates $u = -0.075$ to be an input which will transition to a safe state under any disturbances. However, the system under disturbances $W_1 = 0.2$ and $W_2 = 0.3$ will lead to the state $(0.0188, 0.0131, -0.1167)$ that is outside of the winning domain of the controller. In comparison, the winning domain provided by our RSA method is a subset of the one from the model-based method and provides full guarantees of the satisfaction of the specification and correctness of the controller. This guarantee is obtained at the cost of an increased number of samples and a bias term included in our growth bound calculations, which makes the controller more conservative.

As a final point on this case study, our approach assumes the constant L_φ estimated using sampled trajectories is a correct upper bound on the Lipschitz constant. As mentioned in Remark 5, $\theta_1 = L_\varphi \mathbf{1}_{n\times n}$ can also be used directly to construct the abstraction. The growth bound computed by directly applying the Lipschitz constant is simply conservative: large growth bounds result in controllers with smaller winning domains. In this particular case study, the direct use of the upper bound on the Lipschitz constant (without correction) gives a controller that covers only 78.8% of the winning domain obtained from the model-based approach. Our data-driven approach covers 91.2% as reported in Table 4.

6.5. Parameter optimisation

In this subsection, we discuss how a selection of different parameters can affect the sample complexity and conservativeness of our method. We fix the path planning case study with the estimated Lipschitz constant of 1.46. Figs. 10 and 11 illustrate the effect of changing parameters ϵ, β on the number of samples N required for each pair $(\hat{x}, \hat{u})$ in order to compute a growth bound with confidence $(1 - \beta)$. Fig. 10 illustrates the effect of increasing the confidence parameter β on reducing the sample complexity, for a fixed $\epsilon = 0.01$. Fig. 11 shows that for a fixed $\beta = 0.01$, increasing ϵ leads to a rapid drop in N. In both Figs. 10 and 11, the sample complexity increases in the presence of disturbance as the dimension of the sample space becomes larger.

Fig. 12 demonstrate the effect of changing ϵ on the value of the bias term γ that makes the inequalities of the SCP more conservative. The bias term γ increases for larger values of ϵ. Therefore, increasing ϵ can decrease the sample complexity while increasing γ. Finally, it can be observed that the value of γ is larger in the presence of disturbance.

7. Discussion and future work

We proposed a data-driven method for computing finite abstractions of continuous systems with unknown dynamics. Our approach casts the computation of an overapproximation of reachable sets as a robust convex program (RCP). A feasible solution

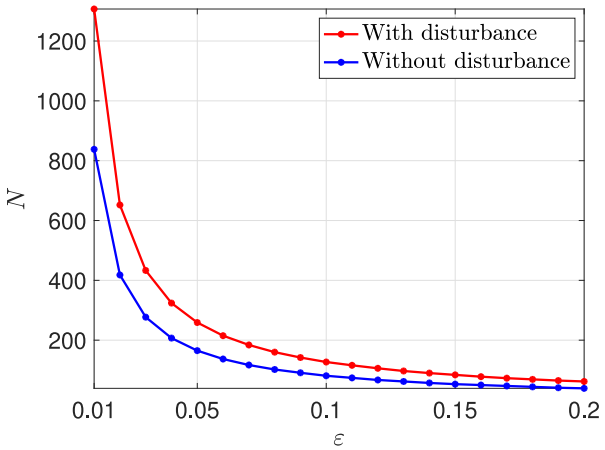


Fig. 11. Required number of samples for our approach as a function of ϵ for a fixed $\beta = 0.01$.

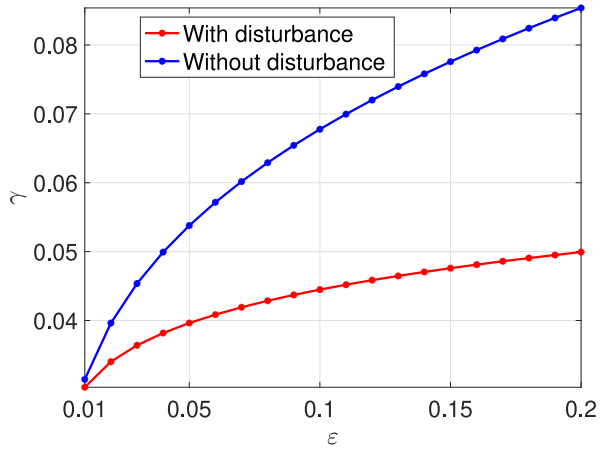


Fig. 12. The bias term γ as a function of ϵ.

for the RCP is then obtained with a given confidence by solving a corresponding scenario convex program (SCP). The SCP does not need the dynamics of the system and requires only a finite set of sample trajectories. We provided a sample complexity result that gives a lower bound on the number of trajectories to achieve a certain confidence.

We guaranteed that with high confidence the computed abstraction is a valid abstraction of the system that overapproximates its behaviours on its entire state space. We showed that our data-driven approach can be embedded into abstraction refinement schemes for designing a controller and enlarging the winning region of the controller with respect to satisfaction of temporal properties. Finally, we evaluated our approach in three case studies.

Our sample complexity result requires knowing a (possibly conservative) bound on the Lipschitz constant of the system. Algorithms founded on the extreme value theory can be utilised to *estimate* the true Lipschitz constant. The estimated values will converge to a true Lipschitz constant when the number of samples used for this estimation goes to infinity. Our sample complexity result for the abstraction and synthesis is valid under the assumption that these estimations give a correct upper bound on the Lipschitz constant. Formal estimation of an upper bound on the Lipschitz constant with finite sample complexity is an interesting research direction, which is out of the scope of this paper. We provided a theorem that shows how such formal estimation methods can be coupled with our approach to give an end-to-end guarantee. We observed that direct use of a bound on the Lipschitz constant without knowing the model of the system gives conservative results and does not provide any opportunity to improve the quality of the computed solution. In contrast to this, the precision of our approach will be improved by increasing the number of samples and performing abstraction refinements.

Our approach has an exponential complexity with respect to the dimension in computing a growth bound. This exponential sample complexity is due to the lack of knowledge of the dynamics and requiring guarantees on satisfaction of robust convex programs. We are not aware of any result in the literature that does not have this exponential complexity while providing the type of guarantees in our paper. On the other hand, our approach is highly parallelisable, and computations can be done individually

for each state–action pair in parallel. The abstraction can also be refined depending on the local computation of a growth bound, thus improving the performance of the model-based approaches depending on the required confidence.

In our approach, we assumed that disturbances are piecewise constant. In the future, we plan to extend our approach by enlarging the class of disturbances beyond piece-wise constant ones (i.e., tackling the issue of infinite dimensional sampling spaces), improve the scalability of the approach by providing more efficient parallel implementation of the approach, and apply it to large case studies that are combinations of differential equations, block diagrams, and lookup tables.

CRediT authorship contribution statement

Milad Kazemi: Conceptualization, Formal analysis, Investigation, Methodology, Writing – original draft. **Rupak Majumdar:** Conceptualization, Funding acquisition, Resources, Supervision. **Mahmoud Salamati:** Conceptualization, Formal analysis, Investigation, Methodology, Writing – original draft. **Sadegh Soudjani:** Conceptualization, Formal analysis, Funding acquisition, Methodology, Resources, Supervision, Writing – review & editing. **Ben Wooding:** Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Sadegh Soudjani reports financial support was provided by UK Research and Innovation. Sadegh Soudjani reports financial support was provided by European Research Council. Sadegh Soudjani reports financial support was provided by European Innovation Council. Rupak Majumdar reports financial support was provided by German Research Foundation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

[1] C. Baier, J. Katoen, Principles of Model Checking, MIT Press, 2008.
[2] C. Belta, B. Yordanov, E.A. Gol, Formal Methods for Discrete-Time Dynamical Systems, Vol. 15, Springer, 2017.
[3] P. Tabuada, Verification and Control of Hybrid Systems: A Symbolic Approach, first ed., Springer Publishing Company, Incorporated, 2009.
[4] A. Girard, G.J. Pappas, Approximation metrics for discrete and continuous systems, IEEE Trans. Automat. Control 52 (5) (2007) 782–798.
[5] R. Majumdar, N. Ozay, A. Schmuck, On abstraction-based controller design with output feedback, in: A.D. Ames, S.A. Seshia, J. Deshmukh (Eds.), HSCC '20: 23rd ACM International Conference on Hybrid Systems: Computation and Control, Sydney, New South Wales, Australia, April 21-24, 2020, ACM, 2020, pp. 15:1–15:11.
[6] S. Samuel, K. Mallik, A. Schmuck, D. Neider, Resilient abstraction-based controller design, in: HSCC '20: 23rd ACM International Conference on Hybrid Systems: Computation and Control, Sydney, New South Wales, Australia, April 21-24, 2020, ACM, 2020, pp. 33:1–33:2.
[7] G. Reissig, A. Weber, M. Rungger, Feedback refinement relations for the synthesis of symbolic controllers, IEEE TAC 62 (4) (2016) 1781–1796.
[8] G. Wood, B. Zhang, Estimation of the Lipschitz constant of a function, J. Global Optim. 8 (1) (1996) 91–103.
[9] T.-W. Weng, H. Zhang, P.-Y. Chen, J. Yi, D. Su, Y. Gao, C.-J. Hsieh, L. Daniel, Evaluating the robustness of neural networks: An extreme value theory approach, in: International Conference on Learning Representations, 2018.
[10] A. Abate, M. Prandini, J. Lygeros, S. Sastry, Probabilistic reachability and safety for controlled discrete time stochastic hybrid systems, Automatica 44 (11) (2008) 2724–2734.
[11] A. Lavaei, S. Soudjani, A. Abate, M. Zamani, Automated verification and synthesis of stochastic hybrid systems: A survey, Automatica 146 (2022) 110617.
[12] I. Mitsioni, P. Tajvar, D. Kragic, J. Tumova, C. Pek, Safe data-driven contact-rich manipulation, in: 2020 IEEE-RAS 20th International Conference on Humanoid Robots, Humanoids, IEEE, 2021, pp. 120–127.
[13] A. Bajcsy, S. Bansal, E. Bronstein, V. Tolani, C.J. Tomlin, An efficient reachability-based framework for provably safe autonomous navigation in unknown environments, in: 2019 IEEE 58th Conference on Decision and Control, CDC, IEEE, 2019, pp. 1758–1765.
[14] F. Djeumou, A.P. Vinod, E. Goubault, S. Putot, U. Topcu, On-the-fly control of unknown systems: From side information to performance guarantees through reachability, IEEE Trans. Automat. Control (2022) 1–16.
[15] K. Grover, F. dos Santos Barbosa, J. Tumova, J. Kretınsky, Semantic Abstraction-Guided Motion Planningfor scLTL Missions in Unknown Environments, in: Robotics: Science and Systems, 2021.
[16] S. Sadraddini, C. Belta, Formal guarantees in data-driven model identification and control synthesis, in: Proceedings of the 21st International Conference on Hybrid Systems: Computation and Control (Part of CPS Week), HSCC 2018, Porto, Portugal, April 11-13, 2018, ACM, 2018, pp. 147–156.
[17] M.H. Cohen, C. Belta, Model-based reinforcement learning for approximate optimal control with temporal logic specifications, in: HSCC '21: 24th ACM International Conference on Hybrid Systems: Computation and Control, Nashville, Tennessee, May 19-21, 2021, ACM, 2021, pp. 12:1–12:11.
[18] C. Fan, B. Qi, S. Mitra, M. Viswanathan, DryVR: Data-driven verification and compositional reasoning for automotive systems, in: Computer Aided Verification - 29th International Conference, CAV 2017, Heidelberg, Germany, July 24-28, 2017, Proceedings, Part I, in: Lecture Notes in Computer Science, vol. 10426, Springer, 2017, pp. 441–461.
[19] Z. Wang, R.M. Jungers, Probabilistic guarantees on the objective value for the scenario approach via sensitivity analysis, in: 2022 IEEE 61st Conference on Decision and Control, CDC, IEEE, 2022, pp. 5668–5673.
[20] G.O. Berger, R.M. Jungers, Z. Wang, Data-driven invariant subspace identification for black-box switched linear systems, in: 2022 IEEE 61st Conference on Decision and Control, CDC, IEEE, 2022, pp. 32–37.
[21] A. Ahmad, C. Belta, R. Tron, Adaptive sampling-based motion planning with control barrier functions, in: 2022 IEEE 61st Conference on Decision and Control, CDC, IEEE, 2022, pp. 4513–4518.

[22] B. Zhong, M. Zamani, M. Caccamo, Synthesizing safety controllers for uncertain linear systems: A direct data-driven approach, in: 2022 IEEE Conference on Control Technology and Applications, CCTA, 2022, pp. 1278–1284.
[23] M.H. Cohen, Z. Serlin, K. Leahy, C. Belta, Temporal logic guided safe model-based reinforcement learning: A hybrid systems approach, Nonlinear Anal. Hybrid Syst. 47 (2023) 101295.
[24] K.-C. Hsu, V. Rubies-Royo, C.J. Tomlin, J.F. Fisac, Safety and liveness guarantees through reach-avoid reinforcement learning, in: Robotics: Science and Systems, 2021.
[25] X. Wang, S. Nair, M. Althoff, Falsification-based robust adversarial reinforcement learning, in: 2020 19th IEEE International Conference on Machine Learning and Applications, ICMLA, IEEE, 2020, pp. 205–212.
[26] C.F. Verdier, N. Kochdumper, M. Althoff, M. Mazo Jr., Formal synthesis of closed-form sampled-data controllers for nonlinear continuous-time systems under STL specifications, Automatica 139 (2022) 110184.
[27] K.C. Kalagarla, R. Jain, P. Nuzzo, Model-free reinforcement learning for optimal control of Markov decision processes under signal temporal logic specifications, in: 2021 60th IEEE Conference on Decision and Control, CDC, 2021, pp. 2252–2257.
[28] D. Sun, S. Jha, C. Fan, Learning certified control using contraction metric, in: Conference on Robot Learning, PMLR, 2021, pp. 1519–1539.
[29] K. Watanabe, N. Renninger, S. Sankaranarayanan, M. Lahijanian, Probabilistic specification learning for planning with safety constraints, in: Intelligent Robots and Systems, IROS, IEEE, 2021, p. TBA.
[30] A. Lavaei, A. Nejati, P. Jagtap, M. Zamani, Formal safety verification of unknown continuous-time systems: A data-driven approach, in: Proceedings of the 24th International Conference on Hybrid Systems: Computation and Control, HSCC '21, Association for Computing Machinery, New York, NY, USA, 2021.
[31] A. Salamati, A. Lavaei, S. Soudjani, M. Zamani, Data-driven verification and synthesis of stochastic systems via barrier certificates, Automatica 159 (111323) (2024) 1–16.
[32] B. Legat, R.M. Jungers, J. Bouchat, Abstraction-based branch and bound approach to Q-learning for hybrid optimal control, in: Learning for Dynamics and Control, PMLR, 2021, pp. 263–274.
[33] A. Makdesi, A. Girard, L. Fribourg, Efficient data-driven abstraction of monotone systems with disturbances, in: 7th IFAC Conference on Analysis and Design of Hybrid Systems, ADHS 2021, Brussels, Belgium, July 7-9, 2021, in: IFAC-PapersOnLine, vol. 54, (no. 5) Elsevier, 2021, pp. 49–54.
[34] T. Badings, L. Romao, A. Abate, D. Parker, H.A. Poonawala, M. Stoelinga, N. Jansen, Robust control for dynamical systems with non-Gaussian noise via formal abstractions, Journal of Artificial Intelligence Research 76 (2023) 341–391.
[35] T.S. Badings, A. Abate, N. Jansen, D. Parker, H.A. Poonawala, M. Stoelinga, Sampling-based robust control of autonomous systems with non-Gaussian noise, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 36, (no. 9) Association for the Advancement of Artificial Intelligence (AAAI), 2022, pp. 9669–9678.
[36] A. Lavaei, S. Soudjani, E. Frazzoli, M. Zamani, Constructing MDP abstractions using data with formal guarantees, IEEE Control Syst. Lett. 7 (2022) 460–465.
[37] A. Virmaux, K. Scaman, Lipschitz regularity of deep neural networks: Analysis and efficient estimation, Adv. Neural Inf. Process. Syst. 31 (2018).
[38] M. Fazlyab, A. Robey, H. Hassani, M. Morari, G. Pappas, Efficient and accurate estimation of Lipschitz constants for deep neural networks, Adv. Neural Inf. Process. Syst. 32 (2019).
[39] M. Jordan, A. Dimakis, Provable Lipschitz certification for generative models, in: International Conference on Machine Learning, PMLR, 2021, pp. 5118–5126.
[40] Y.D. Sergeyev, An information global optimization algorithm with local tuning, SIAM J. Optim. 5 (4) (1995) 858–870.
[41] R. Strongin, K. Barkalov, S. Bevzuk, Acceleration of global search by implementing dual estimates for Lipschitz constant, in: International Conference on Numerical Computations: Theory and Algorithms, Springer, 2019, pp. 478–486.
[42] R. Strongin, On the convergence of an algorithm for finding a global extremum, Eng. Cybern. 11 (1973) 549–555.
[43] J.W. Huang, S.J. Roberts, J.-P. Calliess, On the sample complexity of Lipschitz constant estimation, Trans. Mach. Learn. Res. (2023).
[44] A. Devonport, A. Saoud, M. Arcak, Symbolic abstractions from data: A PAC learning approach, in: 2021 60th IEEE Conference on Decision and Control, CDC, IEEE, 2021, pp. 599–604.
[45] B. Xue, M. Zhang, A. Easwaran, Q. Li, PAC model checking of black-box continuous-time dynamical systems, IEEE Trans. Comput.-Aided Des. Integr. Circuits Syst. 39 (11) (2020) 3944–3955.
[46] G.C. Calafiore, M.C. Campi, The scenario approach to robust control design, IEEE Trans. Automat. Control 51 (5) (2006) 742–753.
[47] S. Soudjani, R. Majumdar, Concentration of measure for chance-constrained optimization, IFAC-PapersOnLine 51 (16) (2018) 277–282.
[48] P.M. Esfahani, T. Sutter, J. Lygeros, Performance bounds for the scenario approach and an extension to a class of non-convex programs, IEEE Trans. Automat. Control 60 (1) (2014) 46–58.
[49] M. Rungger, M. Zamani, SCOTS: A tool for the synthesis of symbolic controllers, in: Proceedings of the 19th International Conference on Hybrid Systems: Computation and Control, 2016, pp. 99–104.
[50] P. Mohajerin Esfahani, T. Sutter, J. Lygeros, Performance bounds for the scenario approach and an extension to a class of non-convex programs, IEEE Trans. Automat. Control 60 (1) (2015) 46–58.
[51] S. Boyd, L. Vandenberghe, Convex Optimization, Cambridge University Press, 2004.
[52] A.M. Mood, Introduction to the Theory of Statistics, McGraw-hill, 1950.
[53] A. Girard, G. Pola, P. Tabuada, Approximately bisimilar symbolic models for incrementally stable switched systems, IEEE Trans. Automat. Control 55 (1) (2009) 116–126.
[54] M. Ma, L. Fan, Implementing consensus based distributed control in power system toolbox, in: 2016 North American Power Symposium, NAPS, 2016, pp. 1–6.
[55] J. Chow, K. Cheung, A toolbox for power system dynamics and control engineering education and research, IEEE Trans. Power Syst. 7 (4) (1992) 1559–1564, http://dx.doi.org/10.1109/59.207380.